

Graph-Aware Deep Reinforcement Learning for Adaptive Capacity Planning in Distributed Personalized Regenerative Medicine Manufacturing Networks

Zhaowei Li^{a,*}, Chin-Yuan Tseng^a

^a*H. Milton Stewart School of Industrial and Systems Engineering, Georgia Institute of Technology, Atlanta, GA, USA*

Abstract

Autologous cell therapy and other personalized regenerative medicine (PRM) products are manufactured on demand across distributed clinical networks, where each product is *perishable*, *identity-bound* to a single patient and non-fungible, and needed only while that patient remains clinically eligible. These characteristics couple capacity, inventory, patient-indexed logistics, and patient outcomes into one networked decision. We formulate distributed PRM capacity planning as a constrained graph Markov decision process and release a reusable simulation testbed with patient-condition dynamics, expiry-aware inventory, geographic transfer delays, identity-preserving pre-manufacturing routing of intact specimen lots between qualified facilities, deterministic return of finished product to the collection site, and sharing of fungible reagents and idle bioreactor capacity. To solve it, we develop an *advantage-filtered residual graph convolutional network-deep deterministic policy gradient* controller (AFR-GCN-DDPG). The study-specific AFR

*Corresponding author.

Email address: zli686@gatech.edu (Zhaowei Li)

label denotes the complete anchor-plus-residual controller, in which a graph convolutional state encoder learns bounded corrections to a transparent two-period look-ahead policy; it is not a new actor-critic backbone. The numerical results currently reported are from earlier configurations in which specimen routing was disabled; they compare controller architectures and are not estimates of the value of routing. In a five-seed demand-prior-drift experiment, with 300 training episodes and 100 paired holdout replications per seed, AFR-GCN-DDPG reduced the total weighted objective by 0.016360% relative to MDL-2, with a two-level 95% confidence interval fully below zero. A matched flat residual DDPG policy also improved on MDL-2, while the graph-versus-flat interval crossed zero, so this replenishment-only experiment supports residual learning but not a graph-specific gain. After extending the residual action to geographic reagent and idle-capacity transfers, a three-seed abrupt-regional-shift experiment reduced the objective by 1.1337% relative to MDL-2 and by 1.1166% relative to a parameter-matched flat policy while improving completion service. A separate conservative online-TD3 screen produced a final graph policy that improved on MDL-2 and matched flat TD3; however, its pooled final-versus-frozen-pretraining interval crossed zero. The evidence therefore supports graph-aware anchored control under geographically structured non-stationarity, while the incremental contribution of online actor-critic updates remains an open attribution question.

Keywords: Graph convolutional networks, Deep reinforcement learning, Actor-critic methods, Capacity planning, Distributed manufacturing, Personalized regenerative medicine, Perishable inventory, Transshipment

1. Introduction

Distributed make-to-order manufacturing networks are increasingly important in engineering systems where products are individualized, production capacity is scarce, and service outcomes depend on time-sensitive sequential decisions. Personalized regenerative medicine (PRM), and autologous cell therapy in particular, is a demanding instance of this class. Each therapy is manufactured from an individual patient’s own biological material, so production can begin only when that patient’s specimen, an available bioreactor, and sufficient consumables are simultaneously present, and the finished product must be returned to the same patient within a limited viability window. Because the product is patient-specific, it cannot be pre-produced, pooled, or substituted, and the opportunity to buffer against uncertainty with inventory is fundamentally limited [1, 2]. Three properties therefore distinguish PRM capacity planning from conventional make-to-order or inventory control. The product is *perishable*, subject to a hard shelf-life and vein-to-vein deadline; it is *identity-bound*, tied one-to-one to a single patient and non-fungible across patients; and its demand is driven by *patient health*, since a patient whose condition deteriorates while waiting may become clinically ineligible before treatment can be delivered. These are not marginal concerns. Autologous CAR-T manufacturing has a turnaround measured in weeks, an estimated 4–7% of patients never receive their product because of manufacturing failure, and in multiple myeloma roughly one quarter of patients die while waitlisted for a BCMA-directed CAR-T production slot [3]. Discrete-event simulation links longer waits directly to survival, with predicted one-year mortality in large B-cell lymphoma rising from about 36% to 76% as vein-to-vein time

26 increases from one to nine months [4]. With the CAR-T market projected to
27 reach tens of billions of US dollars by the early 2030s [5], delivering perish-
28 able, patient-specific product before patients deteriorate is a consequential
29 operational challenge.

30 These properties make capacity planning a coupled, network-level deci-
31 sion problem rather than a collection of independent local ones. Locating
32 production near patients through distributed or point-of-care manufactur-
33 ing shortens fulfillment time, but it also creates spatial dependencies. A
34 shortage at one clinic can sometimes be relieved by relocating idle bioreactor
35 capacity or transshipping consumables from connected facilities, and each
36 such action changes the future state of the whole network [6, 7]. Identity
37 binding constrains ownership and use, not physical location: reagents and
38 idle bioreactor capacity are fungible, whereas each specimen lot and finished
39 product remain attached to one named patient. An intact waiting specimen
40 may therefore be routed to a qualified manufacturing facility without being
41 pooled, split, substituted, or reassigned, and the resulting product must re-
42 turn to that patient’s collection or infusion site. Coordination must account
43 for this patient-indexed logistics pathway as well as fungible-resource shar-
44 ing. The state of such a system is naturally relational, since each facility is a
45 node whose value depends on its neighbors’ queues and inventories and on the
46 links along which shortages, slack, specimens, and disruptions propagate. A
47 flat feature vector encodes local conditions but discards this topology, which
48 motivates a graph-structured state representation as a matter of substance
49 rather than convenience.

50 Reinforcement learning (RL) is a natural fit because the problem is se-

51 quential. An action taken in one period alters future inventories, capacity,
 52 waiting specimens, and service outcomes. Continuous-control actor–critic
 53 methods, including deep deterministic policy gradient (DDPG) [8], twin de-
 54 layed DDPG (TD3) [9], soft actor–critic (SAC) [10], and proximal policy op-
 55 timization (PPO) [11], are standard tools for such real-valued control. Graph
 56 convolutional networks (GCNs) [12] provide a permutation-equivariant rela-
 57 tional inductive bias for encoding networked state. This machinery is by now
 58 well established, and a substantial literature already couples graph learning
 59 with reinforcement learning for supply-chain and resource-allocation prob-
 60 lems [13, 14, 15, 16, 17]. We therefore do not claim novelty in combining
 61 graph representation with reinforcement learning.

62 What that literature does not model is the PRM problem itself. Prior
 63 graph-and-RL studies for networked operations concentrate on demand fore-
 64 casting, inventory-only ordering, or generic enterprise resource allocation,
 65 and they represent demand as fungible, abstract requests [13, 15, 18, 19,
 66 20]; none model a product that expires, production that must return to the
 67 same patient, or demand generated by patient deterioration while waiting.
 68 Reinforcement learning has been applied within cell-therapy manufacturing,
 69 but at the level of a single bioprocess rather than a network of clinics [2],
 70 and recent evidence cautions that learned policies do not uniformly domi-
 71 nate well-tuned operational heuristics, particularly under perishability and
 72 non-stationarity [21]. The gap we address is thus a problem gap rather than
 73 a method gap, namely perishable, identity-bound, patient-condition-driven
 74 capacity planning across a clinic network, together with a decision framework
 75 and simulation testbed built specifically for it. To the best of our knowledge

76 no public benchmark exists for this setting, so the environment is itself a
77 contribution.

78 We organize the study around five research questions. **(RQ1)** With
79 patient-indexed specimen routing held admissible across policies, does a
80 graph representation of the clinic network improve coordinated specimen,
81 capacity, reagent, and replenishment decisions over flat-state control, and
82 under which supply-disruption and forecast-error conditions is any improve-
83 ment largest? **(RQ2)** Which actor-critic backbone (DDPG, TD3, SAC,
84 or PPO) is best suited to this constrained continuous-control problem, and
85 is DDPG alone adequate? **(RQ3)** Does explicitly modeling patient condi-
86 tion and product expiry change the optimal policy and expose failures of
87 condition-blind planners? **(RQ4)** Does a condition- and expiry-aware tem-
88 poral encoder, which retains memory of deterioration trajectories, improve
89 on a static graph encoder? **(RQ5)** Do learned policies outperform strong
90 look-ahead heuristics, not only weak myopic rules, under supply disruption,
91 demand forecast error, and patient-condition stress?

92 To study these questions we develop, first, a simulation of a multi-site
93 autologous cell-therapy network in which patient health conditions, patient-
94 indexed specimen routing, and product and material expiry are first-class
95 drivers of demand and feasibility, and second, an advantage-filtered residual
96 GCN-DDPG (AFR-GCN-DDPG) controller that coordinates specimen rout-
97 ing, capacity, reagent, and replenishment decisions across the network. AFR
98 is our shorthand for the complete controller design: a transparent MDL-2 an-
99 chor, a graph-aware bounded residual actor, evidence-filtered teacher targets,
100 and independently calibrated deployment safeguards. It does not replace or

101 rename DDPG. The graph component encodes each facility’s operational
 102 and patient-condition state together with information from connected facil-
 103 ities into network-aware embeddings. A deterministic policy-gradient actor
 104 then learns a bounded, state-dependent correction to the anchor policy, and
 105 a feasibility projection converts the corrected action into implementable con-
 106 trols. Advantage-filtered distillation retains a nonzero correction target only
 107 when common-random-number look-ahead predicts an improvement over the
 108 anchor; it is a teacher-data construction and regularization step, not a sec-
 109 ond algorithm or deployed policy. Checkpoint and residual-scale calibration
 110 provide a deployment safeguard. DDPG is the canonical proposed instantia-
 111 tion and provides continuity with prior PRM capacity-planning work; graph-
 112 aware TD3, SAC, and PPO variants serve as controlled stability and back-
 113 bone comparisons. We adopt a statistically disciplined evaluation protocol
 114 (multiple random seeds, paired Monte Carlo outcomes, and two-level con-
 115 fidence intervals), because single-seed, mean-only reporting can make deep
 116 reinforcement learning results appear stronger and more stable than they
 117 are [22, 23]. Consistent with the clinical stakes of the problem, we report
 118 patient-service outcomes such as eligibility and waiting alongside operating
 119 cost, since a lower average cost can mask clinically unacceptable delay.

120 The main contributions of this study are as follows.

- 121 • We formulate distributed capacity planning for *perishable, identity-*
 122 *bound, patient-condition-driven* PRM manufacturing as a constrained
 123 networked sequential decision problem, in which patient deterioration
 124 and product and material expiry are explicit determinants of demand,
 125 feasibility, and cost.

- 126 • We develop a simulation environment realizing this formulation for a
127 multi-clinic autologous cell-therapy network, with patient-condition dy-
128 namics, expiry-aware inventory, patient-indexed and identity-preserving
129 pre-manufacturing specimen routing, finished-product return, and fungible-
130 resource sharing. Routing is an operational capability of the model
131 rather than a claimed methodological contribution.
- 132 • We propose AFR-GCN-DDPG, an advantage-filtered residual graph
133 controller that learns bounded corrections to a transparent look-ahead
134 anchor under demand and disruption uncertainty, with persistent teacher
135 regularization and independently calibrated deployment safeguards.
- 136 • We establish a paired multi-seed evaluation protocol for comparisons
137 with strong look-ahead and isolated-facility heuristics, flat-state rein-
138 forcement learning, and graph-aware backbone variants, and report a
139 five-seed DDPG confirmation, a parameter-matched graph-attribution
140 study, and a conservative online-TD3 screen with two-level bootstrap
141 confidence intervals.
- 142 • We draw design implications for intelligent decision support in dis-
143 tributed, patient-specific healthcare manufacturing and related time-
144 critical, make-to-order engineering systems.

145 The remainder of this paper is organized as follows. Section 2 formu-
146 lates the distributed PRM capacity planning problem, extended with patient-
147 condition states and product and material expiry. Section 3 reviews personal-
148 ized regenerative medicine operations, reinforcement- and graph-learning for

149 networked operational decisions, and the method foundations of continuous-
150 control reinforcement learning and graph neural networks, closing with the
151 design implications that motivate our approach. Section 4 presents the graph-
152 aware deep reinforcement learning framework. Section 5 describes the com-
153 putational experiments and results. Section 6 concludes the study and iden-
154 tifies future research directions.

155 **2. Distributed Network Manufacturing and Capacity Planning Prob-** 156 **lem**

157 In this section, we formulate the capacity planning problem for distributed
158 manufacturing systems as a production simulation model based on a Markov
159 decision process (MDP). Suppose the distributed manufacturing system con-
160 sists of N manufacturing facilities that produce one type of make-to-order
161 PRM product. Each manufacturing facility has its own production capacity,
162 including bioreactors and reagent inventory, and can purchase reagents from
163 source material suppliers. Facilities receive patient-indexed specimens and
164 initiate production when the matched specimen, an available bioreactor, and
165 sufficient reagents are co-located. Reagent transshipment and idle-bioreactor
166 relocation are allowed among qualified facilities. In addition, one intact wait-
167 ing specimen lot may be routed before manufacturing without changing the
168 patient-specimen association; pooling, splitting, substitution, cross-patient
169 use, and in-process transfer remain prohibited, and finished product is re-
170 turned to the same patient’s collection site (Section 2.3). Demand arrivals
171 and supplier disruptions are modeled as stochastic processes, implying that
172 local supply may be either insufficient or excessive relative to local demand

Table 1: Notation used in the distributed manufacturing and capacity planning model.

Sets and parameters	
Γ	Planning horizon indexed by t .
T	Manufacturing time for one PRM product, in periods, where $T < \Gamma$.
N	Set of manufacturing facilities indexed by n .
$\mathcal{E}_S$	Qualified facility pairs along which an intact waiting specimen lot may be routed before manufacturing.
m_t^n	Products that begin manufacturing at facility n at epoch t .
d_t^n	Demand arrivals at facility n in epoch t .
State variables	
s_t^n	Patient-indexed specimens waiting for manufacturing at their current facility n at epoch t .
r_t^n	Available reagent units at facility n at epoch t .
$\mathbf{b}_t^n$	Bioreactor work-in-process vector $(b_t^{n,1}, b_t^{n,2}, \dots, b_t^{n,T})$, where $b_t^{n,\tau}$ denotes the number of bioreactors at facility n with τ periods remaining at epoch t .
P_t^n	Maximum reagent purchase quantity for facility n for epoch $t + 1$.
Decision variables	
w_t^n	Executed integer net flow of intact waiting specimen lots into (positive) or out of (negative) facility n at epoch t , with identity and network conservation enforced (Section 2.3).
e_t^n	Reagent units added to or subtracted from facility n at epoch t and received or removed before epoch $t + 1$.
q_t^n	Idle bioreactors added to or subtracted from facility n at epoch t and received or removed before epoch $t + 1$.
p_t^n	Reagent purchase quantity for facility n for epoch $t + 1$.
Patient-condition and perishability	
H_j	Latent health index of patient j , drawn $H_j \sim \text{Uniform}(0, 1)$ at enrollment.
τ_j	Deterioration epoch of patient j , drawn $\tau_j \sim \theta \text{Weibull}(\kappa)$.
κ, θ	Shape and scale of the Weibull deterioration-epoch distribution.
α_j	Number of epochs since patient j enrolled, including waiting and manufacturing time.

173 at different decision epochs. The notation used throughout this section is
 174 summarized in Table 1.

175 Figure 1 illustrates the basic structure of the distributed manufacturing
 176 problem using a two-facility example. Each facility maintains a patient–
 177 specimen queue whose members carry a deteriorating survival index, together
 178 with reagent inventory, bioreactor capacity, and perishable finished product.
 179 At each decision epoch, condition-driven patient demand and disruption-
 180 prone reagent replenishment arrive under uncertainty. When one facility
 181 experiences excess demand, insufficient reagents, or limited production ca-
 182 pacity, the network may transship reagents, relocate idle bioreactors, or route
 183 an intact waiting specimen to a qualified manufacturing facility. The latter
 184 is patient-indexed logistics rather than sharing: identity, ownership, and the
 185 final infusion destination never change. This structure motivates the MDP
 186 formulation, where the state captures facility-level demand, patient condi-
 187 tion, material location and transit, inventory, and capacity, and the action
 188 coordinates both patient-indexed routing and fungible-resource decisions.

189 For the N facilities, T epochs of manufacturing time are required to
 190 produce one of the PRM products, and the planning horizon consists of Γ
 191 epochs where $\Gamma > T$.

192 At epoch $t \in \{1, 2, \dots, \Gamma\}$, the production status of the n -th facility for
 193 $n = 1, \dots, N$ is expressed as a state vector $\mathbf{x}_t^n$, where

$$\mathbf{x}_t^n = (s_t^n, r_t^n, \mathbf{b}_t^n, P_t^n), \quad (1)$$

194 Based on the state variables at epoch t , the facilities need to take action
 195 to adjust their production capacity for epoch $t + 1$. We define the action

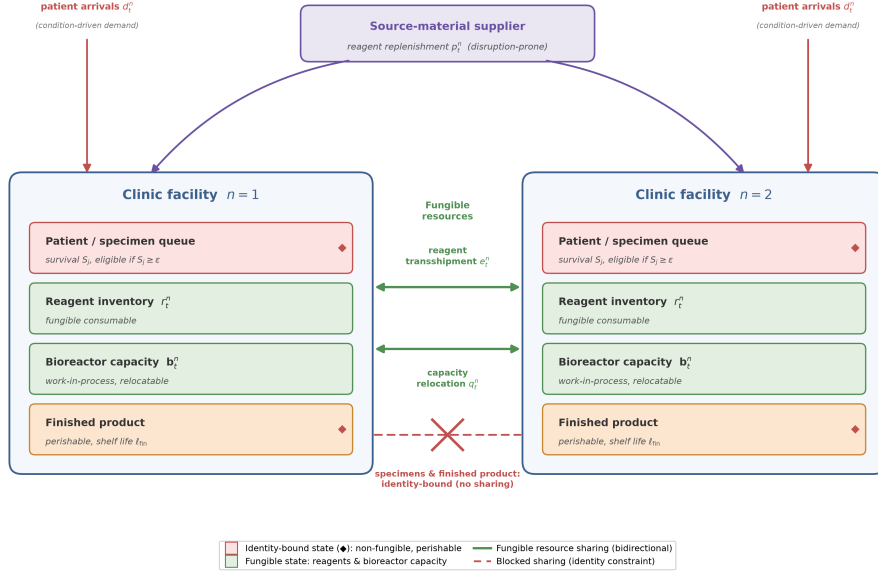


Figure 1: Illustration of a two-facility distributed manufacturing system for personalized regenerative medicine. Each facility holds a patient–specimen queue whose members carry a deteriorating survival index S_j (eligible while $S_j \geq \epsilon$), reagent inventory r_t^n , bioreactor capacity b_t^n , and perishable finished product. Condition-driven patient demand d_t^n and disruption-prone reagent replenishment p_t^n arrive under uncertainty. Reagents and idle bioreactor capacity are fungible and can be rebalanced through transshipment e_t^n and relocation q_t^n (solid green links). The crossed red link denotes prohibited pooling, substitution, or cross-patient reassignment of specimens and products; it does not prohibit identity-preserving transport of one intact waiting specimen lot to a qualified manufacturing facility or return of its product to the same patient.

196 $\mathbf{a}_t^n$ as a vector that consists of the amount of capacity sharing and reagent
 197 replenishment. The action is decided at the beginning of the production
 198 period t and executed before the start of $t + 1$.

$$\mathbf{a}_t^n = (w_t^n, c_t^n, q_t^n, p_t^n), \quad (2)$$

199 The goal of capacity planning is to minimize total production cost by
 200 dynamically selecting the actions from Equation (2) for all facilities, denoted
 201 by $\mathbf{A}_t = (\mathbf{a}_t^1, \dots, \mathbf{a}_t^N)$, for each epoch $t \in \Gamma$. We define $V(\mathbf{X}_t)$ as the total
 202 future cost of the selecting actions $\mathbf{A}_t$ when the states from all facilities,
 203 denoted by $\mathbf{X}_t = (\mathbf{x}_t^1, \dots, \mathbf{x}_t^N)$, are observed at epoch $t \in \Gamma$. The objective
 204 function can be expressed as a dynamic programming problem:

$$V(\mathbf{X}_t) = \min_{\mathbf{A}_t} E[c(\mathbf{A}_t; \mathbf{X}_t) + \gamma V(\mathbf{X}_{t+1})], \quad (3)$$

205 which is useful for finding the optimal action $\mathbf{A}_t$ for each state $\mathbf{X}_t$ at
 206 each epoch $t \in \Gamma$, where $c(\mathbf{A}_t; \mathbf{X}_t)$ is the immediate cost at time t , and
 207 $\gamma \in [0, 1]$ is a discount factor for balancing the impact of future costs in
 208 deciding actions. The choice of γ depends on the user's goal; if γ is closer
 209 to 1 (to 0), then it means the user cares more about long (short) term
 210 cost; if $\gamma = 0$, then only the cost for current epoch is considered. Equation
 211 (3) generalizes the optimality equation for the capacity planning problem in
 212 Li(2021) [6] in that cost functions $\mathbf{c}$ and $\mathbf{V}$ and the cumulative distribution
 213 function for the expectation operator can be unknown.

214 Equations (1)–(3) describe the base operational MDP shared with generic
 215 networked replenishment problems. The remainder of this section extends it
 216 along the three dimensions that distinguish PRM capacity planning, namely

217 the deteriorating clinical condition of the patients awaiting treatment, the
 218 hard expiry of biological source material and finished product, and the autol-
 219 ogous identity constraint that forbids pooling of patient-specific specimens.
 220 Section 4 then develops a graph-aware DRL approach that learns the cost
 221 function $\mathbf{c}$, the value function $\mathbf{V}$, and the expectation operator directly from
 222 simulated experience.

223 *2.1. Patient condition and treatment eligibility*

224 In generic capacity planning, a unit of unmet demand is deferred, back-
 225 ordered, or lost at a fixed penalty, and its value does not change while it
 226 waits. PRM demand behaves differently, because each arrival is a patient
 227 whose clinical condition deteriorates throughout the vein-to-vein pathway.
 228 A specimen that waits or travels too long, or a therapy whose matched pa-
 229 tient deteriorates during manufacturing or product return, may no longer
 230 produce a deliverable treatment. We therefore attach an explicit condition
 231 process to every enrolled patient from queue entry through specimen transit,
 232 manufacturing, and product return, adapting the real-time patient-health
 233 modeling of Tseng et al. [24] to a distributed capacity-planning setting.

234 When a patient j enrolls at facility n , we draw a latent health index $H_j \sim$
 235 $\text{Uniform}(0, 1)$ and a deterioration epoch $\tau_j \sim \theta \text{Weibull}(\kappa)$ with shape κ and
 236 scale θ ; H_j captures the heterogeneous baseline robustness across patients,
 237 while τ_j marks the onset of accelerated decline. Let α_j be the number of
 238 epochs since enrollment, including both queueing and manufacturing time.
 239 The patient’s survival index $S_j \in [0, 1]$ (a normalized proxy for the likelihood
 240 that the patient remains treatable) evolves as

$$S_j(\alpha_j) = \ell(\alpha_j) + H_j(u(\alpha_j) - \ell(\alpha_j)), \quad (4)$$

241 where $u(\alpha) = \exp(-\lambda_H g(\alpha))$ and $\ell(\alpha) = \exp(-\lambda_F g(\alpha))$ are the survival
 242 trajectories of an ideally healthy and a maximally frail patient, with decay
 243 rates $\lambda_H < \lambda_F$, so that H_j interpolates each patient between the two bounds.
 244 The effective exposure time

$$g(\alpha) = \begin{cases} \alpha, & \alpha \leq \tau_j, \\ \tau_j + \rho(\alpha - \tau_j), & \alpha > \tau_j, \end{cases} \quad (5)$$

245 accelerates by a factor $\rho > 1$ once the patient passes the deterioration epoch
 246 τ_j , reflecting the clinically observed transition to rapid decline. A patient
 247 remains eligible for treatment only while $S_j \geq \epsilon$. Before production starts,
 248 crossing the threshold removes the patient and specimen from either the
 249 waiting or transit state. After production starts, the same condition process
 250 continues through manufacturing and any modeled product-return epochs.
 251 If the matched patient becomes ineligible before completion, the in-process
 252 therapy is discarded and the occupied bioreactor is cleaned during that epoch
 253 before returning to the idle pool at the next decision epoch. Treatment ser-
 254 vice is recorded only when the finished product reaches and is assigned to the
 255 original patient, not when production starts. Because condition erodes mono-
 256 tonically in expectation and accelerates after τ_j , waiting, specimen transit,
 257 manufacturing, and product-return delay are not neutral buffering decisions.
 258 Parameter values are reported in Section 5.

259 2.2. Perishability of materials and finished product

260 Both the biological source material and the finished product are per-
 261 ishable and identity-specific, so they cannot be stockpiled against future de-
 262 mand. A waiting or in-transit specimen that is not put into production within
 263 ℓ_{mat} epochs of enrollment expires, so the patient is lost and the reserved ma-
 264 terial is wasted. Finished product that is not delivered within ℓ_{fin} epochs of
 265 completion likewise expires and is discarded. We track, per facility n and
 266 epoch t , patients lost to waiting-stage ineligibility $\phi_t^{n,\text{wait}}$, manufacturing-
 267 stage ineligibility $\phi_t^{n,\text{mfg}}$, return-stage ineligibility $\phi_t^{n,\text{ret}}$, local specimen ex-
 268 piry $\phi_t^{n,\text{exp}}$, transit ineligibility $\phi_t^{n,\text{tr-inel}}$, transit expiry $\phi_t^{n,\text{tr-exp}}$, and modeled
 269 transit loss $\phi_t^{n,\text{tr-loss}}$, together with finished units lost to expiry $\psi_t^{n,\text{fin}}$. A
 270 manufacturing-stage ineligibility also creates one discarded in-process ther-
 271 apy. It is convenient to collect the total patients lost and total material
 272 wasted at each facility,

$$\begin{aligned}
 \phi_t^n &= \phi_t^{n,\text{wait}} + \phi_t^{n,\text{mfg}} + \phi_t^{n,\text{ret}} + \phi_t^{n,\text{exp}} \\
 &\quad + \phi_t^{n,\text{tr-inel}} + \phi_t^{n,\text{tr-exp}} + \phi_t^{n,\text{tr-loss}}, \\
 \psi_t^n &= \phi_t^{n,\text{exp}} + \phi_t^{n,\text{mfg}} + \phi_t^{n,\text{tr-exp}} \\
 &\quad + \phi_t^{n,\text{tr-loss}} + \psi_t^{n,\text{fin}},
 \end{aligned} \tag{6}$$

273 where an expired or lost specimen contributes both a lost patient and a unit of
 274 wasted material. To signal impending losses before they occur, we also count
 275 the *at-risk unserved* patients whose survival has fallen within an urgency
 276 margin δ of the eligibility boundary, $v_t^n = |\{j \text{ waiting at } n : S_j < \epsilon + \delta\}|$.

277 *2.3. Autologous identity constraint and admissible transfers*

278 The autologous nature of PRM imposes a constraint with no analog in
279 commodity supply chains. A specimen and the product manufactured from it
280 are bound to one patient and cannot be substituted, pooled, split, or delivered
281 to anyone else. Identity preservation does not, however, require the source
282 material to remain at its collection site. We therefore allow *patient-indexed,*
283 *identity-preserving, pre-manufacturing specimen routing*: only a complete
284 lot in the waiting state may move, only along a qualified edge in $\mathcal{E}_S$, and at
285 most once by a direct route before manufacturing begins. Multi-hop routing,
286 cycles, in-production transfer, cross-patient consumption, and product
287 reassignment are prohibited.

288 The aggregate decision w_t^n records the executed integer net flow produced
289 by a patient-level routing executor. The executor pairs facilities with opposite
290 signed requests, selects eligible waiting identities deterministically, and
291 conserves lots so that $\sum_n w_t^n = 0$. Each selected patient's immutable identity
292 record retains the collection facility, current material location, manufacturing
293 facility, and transfer history. The main routing setting dispatches at epoch
294 t , advances condition and specimen age once during one transit epoch, and
295 makes the lot available before the epoch $t + 1$ production decision. After
296 off-site manufacturing, the finished product remains assigned to the same
297 patient and is returned to that patient's collection or infusion facility. Thus,
298 routing changes manufacturing location without creating unrestricted specimen
299 sharing.

300 2.4. Extended cost and the constrained graph MDP

301 The immediate cost augments the operational base cost c_{op} (covering
302 reagent purchase, holding, and shortage; bioreactor holding and shortage;
303 and reagent and capacity transfer) with three condition- and perishability-
304 driven terms:

$$c(\mathbf{A}_t; \mathbf{X}_t) = c_{\text{op}}(\mathbf{A}_t; \mathbf{X}_t) + \omega_{\text{lost}} \sum_{n \in N} \phi_t^n + \omega_{\text{exp}} \sum_{n \in N} \psi_t^n + \omega_{\text{urg}} \sum_{n \in N} v_t^n, \quad (7)$$

305 with weights ordered $\omega_{\text{lost}} \gg \omega_{\text{exp}} > \omega_{\text{urg}}$ to encode the clinical priority, in
306 which losing a patient dominates wasting material, which in turn dominates
307 the shaping penalty for letting patients drift toward the eligibility boundary.
308 The urgency term v_t^n steers the controller to act pre-emptively (starting
309 production or relocating capacity) before losses are actually incurred.

310 For clarity, we distinguish three quantities throughout the paper. For a
311 policy π , the expected total weighted objective is

$$J_{\text{tot}}^\pi = \mathbb{E}_\pi \left[\sum_{t=0}^{\Gamma-1} c(\mathbf{A}_t; \mathbf{X}_t) \right] = J_{\text{op}}^\pi + J_{\text{pat}}^\pi, \quad (8)$$

312 where J_{op}^π accumulates reagent purchase, holding, and shortage; bioreactor
313 holding and shortage; and reagent and capacity transfer, whereas J_{pat}^π accu-
314 mulates the weighted patient-loss, expiry, and urgency terms in Equation (7).
315 The total objective is the primary optimization and evaluation criterion. The
316 two component groups are reported separately to show which decisions gen-
317 erate an improvement.

318 All three quantities are reported in *modeled weighted-objective units*. Resource-
319 price coefficients can be interpreted as monetary equivalents under the adopted

calibration, but the patient-loss, expiry, and urgency weights are penalty or shadow-value parameters rather than observed accounting expenditures. Consequently, an absolute difference in J_{tot} is not presented as realized cash savings without an external economic calibration. Some large components are driven partly by exogenous demand and patient deterioration and by structural constraints such as finite capacity and lead times. They are therefore only partly controllable, not “fixed” or wholly unavoidable.

For an anchor policy π_{anc} and candidate π , the headline relative improvement retains the complete objective as its denominator,

$$R_{\text{tot}} = \frac{J_{\text{tot}}^{\pi_{\text{anc}}} - J_{\text{tot}}^{\pi}}{J_{\text{tot}}^{\pi_{\text{anc}}}}. \quad (9)$$

We accompany it with absolute paired differences, component-specific relative changes, and clinical outcomes. When a valid oracle or lower-cost reference π_{ref} is available, we additionally report the fraction of anchor headroom captured,

$$R_{\text{headroom}} = \frac{J_{\text{tot}}^{\pi_{\text{anc}}} - J_{\text{tot}}^{\pi}}{J_{\text{tot}}^{\pi_{\text{anc}}} - J_{\text{tot}}^{\pi_{\text{ref}}}}. \quad (10)$$

This decomposition prevents a large patient-penalty or capacity-shortage component from obscuring a meaningful change in a directly affected operating component, while avoiding a post hoc replacement of the primary denominator.

To act on this cost, the controller needs visibility into the condition of each waiting queue. We therefore extend the facility state of Equation (1) with a condition summary $\boldsymbol{\sigma}_t^n$,

$$\tilde{\mathbf{x}}_t^n = (s_t^n, r_t^n, \mathbf{b}_t^n, P_t^n, \boldsymbol{\sigma}_t^n), \quad (11)$$

where σ_t^n contains the waiting count, waiting mean survival, near-expiry count, waiting survival-band histogram, specimens in transit to the facility, their mean survival, transferred specimens waiting at the facility, route-eligible waiting specimens, in-production count, in-production mean survival, and in-production at-risk count. The controller can therefore anticipate queue, transit, and manufacturing risk rather than merely react to aggregate demand.

Finally, we cast the extended problem as a *constrained graph MDP*. The facilities and their admissible information, reagent, capacity, and specimen-logistics links form a graph $\mathcal{G} = (N, \mathcal{E})$, with the qualified specimen edge set $\mathcal{E}_S \subseteq \mathcal{E}$ enforced separately from fungible-resource feasibility. The global state $\tilde{\mathbf{X}}_t = (\tilde{\mathbf{x}}_t^1, \dots, \tilde{\mathbf{x}}_t^N)$ is a signal over the nodes of $\mathcal{G}$, and the action $\mathbf{A}_t$ assigns per-node purchases, fungible-resource transfers, and conserved patient-indexed specimen-routing requests. The optimality equation (3) carries over verbatim with the extended state $\tilde{\mathbf{X}}_t$ and cost (7), but the value function must now trade current operating and logistics cost against the future risk of waiting, transit, patient loss, and expiry. This constrained graph MDP (perishable, identity-bound, and patient-condition-driven) is the problem the remainder of the paper addresses, and its graph structure is precisely what the method of Section 4 is designed to exploit.

3. Literature Review

We review three bodies of work that our approach draws together, namely the operations of personalized regenerative medicine (PRM) and decentralized manufacturing; reinforcement learning and graph learning for networked

operational decisions; and the underlying method foundations of continuous-control reinforcement learning and graph neural networks. Rather than surveying each exhaustively, we read them as an argument in which each stream supplies part of what a distributed PRM capacity-planning controller needs, none supplies all of it, and together they imply a small set of design choices that we make explicit at the end of the section. Throughout, we treat the learning machinery as established and concede that graph representation and reinforcement learning have already been combined for supply-chain problems; what remains open is the PRM problem itself, one that is perishable, identity-bound, and patient-condition-driven.

3.1. *Personalized regenerative medicine and decentralized manufacturing*

PRM supply chains differ substantially from conventional make-to-stock manufacturing systems because products are individualized, time-sensitive, and tightly coupled with patient-specific clinical pathways. In autologous cell therapy, patient material must be collected, transported, manufactured, quality-tested, and returned for administration, and manufacturing can begin only when a specimen, consumables, and production capacity are jointly available. These characteristics make fulfillment time, service reliability, and resource utilization central operational measures. Wang et al. [1] used simulation to compare centralized and point-of-care autologous cell therapy supply chain strategies and evaluated cost, fulfillment time, service level, and resource utilization. Li and White [6] further studied decentralized autologous cell therapy manufacturing networks and showed that transshipment and relocatable capacity can reduce the cost of resilience under supplier disruption. Most directly relevant to our patient-condition modeling, Tseng et al. [24]

integrated real-time patient health data into an autologous cell-therapy simulation and showed that representing patient condition, rather than treating demand as static, changes fulfillment and eligibility outcomes. We adopt this patient-condition perspective as a first-class driver of demand and feasibility, but embed it within a networked capacity-planning formulation solved by graph-aware learned control, rather than evaluating fixed policies in a simulation study alone.

These studies establish that PRM network design and inter-facility coordination are first-order engineering decisions rather than implementation details. Related work on cell therapy manufacturing process control also highlights the potential of reinforcement learning in uncertain, nonlinear, and partially observed manufacturing environments [2]. Together, this literature motivates a data-driven decision-support framework that can represent both the temporal dynamics of patient-specific manufacturing and the spatial dependencies created by distributed facilities.

3.2. Reinforcement learning for supply chain and manufacturing decisions

Reinforcement learning has been widely studied for sequential supply chain and manufacturing decisions. Early work applied RL to inventory replenishment in multi-echelon supply chains and perishable inventory systems [25, 26]. More recent studies have used deep reinforcement learning to handle high-dimensional state spaces, stochastic demand, and simulation-based decision environments. For example, Oroojlooyjadid et al. [27] developed a deep Q-network approach for the beer game, demonstrating that DRL can learn replenishment policies in decentralized, partially observed supply chain settings.

Recent supply chain RL studies have moved toward more realistic constraints and engineering settings. Bermudez et al. [19] proposed distributional constrained reinforcement learning for multi-period supply chain optimization with production and inventory constraints. Eisenach et al. [20] studied shared-resource capacity control in inventory management using a neural coordination mechanism and a reinforcement learning buying policy. Stranieri et al. [21] benchmarked classical and deep reinforcement learning inventory policies for pharmaceutical supply chains with perishability and non-stationary demand. These studies support the use of simulation-trained RL policies for uncertain operational control, but most focus on replenishment or inventory management rather than the joint capacity, reagent, and replenishment decisions (under a perishable, identity-bound demand process) that arise in distributed PRM manufacturing.

3.3. *Structure-informed, residual, and baseline-relative reinforcement learning*

Inventory-control evidence does not support treating generic end-to-end DRL as automatically superior to established operating rules. Gijsbrechts et al. [28] found that carefully tuned DRL can match state-of-the-art policies across lost-sales, dual-sourcing, and multi-echelon problems, while the invited roadmap of Boute et al. [29] argues that learning methods should exploit the structural policy knowledge accumulated in inventory research. De Moor et al. [30] made this principle operational by using strong perishable-inventory heuristics as teacher policies for reward shaping, improving training speed, stability, and, in suitable instances, performance relative to the teacher. Hybrid decomposition offers another route: Stranieri et al. [31] assigned pro-

439 duction decisions to DRL and shipping decisions to stochastic programming,
 440 obtaining more reliable performance than pure DRL. Non-stationary-demand
 441 experiments further show the value of conditioning policies on forecast infor-
 442 mation rather than assuming a fixed arrival law [32].

443 Residual reinforcement learning provides the most direct methodologi-
 444 cal foundation for our controller. The residual formulation decomposes the
 445 executed action into a competent base-controller action and a learned cor-
 446 rection, reducing the part of the control law that must be discovered from
 447 experience [33]. Staessens et al. [34] showed in an industrial-control setting
 448 that bounding the correction relative to a conventional controller can pre-
 449 serve its stabilizing role while adapting to uncertain operating conditions.
 450 Baseline-relative policy-improvement methods likewise motivate accepting a
 451 learned update only when there is credible evidence of improvement over
 452 the incumbent policy [35]. In stochastic inventory control, Deep Controlled
 453 Learning combines approximate policy iteration with common random num-
 454 bers and statistically efficient candidate selection to overcome the difficulty
 455 generic DRL has in beating strong heuristics [36]. These ideas motivate our
 456 MDL-2 anchor, bounded graph residual, common-random-number advantage
 457 filtering, and held-out deployment calibration. Our validation fallback is an
 458 empirical deployment safeguard, however, and should not be interpreted as
 459 inheriting the formal monotonic-improvement or closed-loop stability guar-
 460 antees proved under the assumptions of those methods.

461 3.4. *Deep reinforcement learning for regenerative medicine capacity planning*

462 The closest antecedent to this work is Tseng et al. [7], who developed a
 463 deep reinforcement learning approach for dynamic capacity planning in de-

464 centralized regenerative medicine supply chains. Their study demonstrated
 465 that simulation-trained policies can adapt to demand and supply uncertainty
 466 without requiring exact prior knowledge of the underlying stochastic pro-
 467 cesses. This is directly aligned with the objective of using data-driven policy
 468 learning for PRM capacity planning.

469 The present study builds on this line of work by adding an explicit graph
 470 representation layer to encode inter-facility relationships. In decentralized
 471 PRM networks, facilities are linked through reagent transshipment, idle-
 472 bioreactor relocation, and qualified patient-indexed specimen logistics, while
 473 every specimen and finished product remains bound to one patient. There-
 474 fore, a facility’s decision depends not only on its own local state but also
 475 on the states, route-eligible queues, transit pipeline, and fungible-resource
 476 conditions of connected facilities. A flat vectorized state representation can
 477 include all facility variables, but it does not explicitly exploit the graph struc-
 478 ture that governs how information, material, and resources propagate across
 479 the network. This motivates integrating graph convolutional representation
 480 learning with continuous-control actor–critic reinforcement learning.

481 *3.5. Multi-agent reinforcement learning and graph learning for networked* 482 *supply chains*

483 Recent multi-agent reinforcement learning studies are relevant because
 484 distributed manufacturing networks naturally involve decentralized infor-
 485 mation and coordinated actions. Mousa et al. [18] formulated decentral-
 486 ized inventory control as a multi-agent problem and studied centralized-
 487 training/decentralized-execution policies under local information constraints.
 488 Leluc et al. [37] similarly demonstrated the promise of cooperative multi-

489 agent reinforcement learning for inventory management under stochastic de-
490 mand and lead times. These studies show that decentralization is not only
491 an application feature but also an algorithmic design dimension.

492 In parallel, graph neural networks have emerged as a natural repre-
493 sentation tool for supply chains and other networked operational systems.
494 Chang et al. [14] learned production functions on supply-chain graphs us-
495 ing temporal graph neural networks and an inventory module, showing the
496 value of explicit relational structure for supply-chain inference and fore-
497 casting. Ahn et al. [15] used attention-based graph neural networks for
498 probabilistic supply and inventory prediction in enterprise supply-chain net-
499 works, while Ahn et al. [16] combined graph neural networks, offline re-
500 inforcement learning, and policy simulation for dynamic supply planning.
501 Kotecha and del Rio Chanona [13] provide the closest graph-control an-
502 tecedent: their GNN-MARL framework dynamically parameterizes heuris-
503 tic inventory policies and uses graph structure to coordinate decentralized
504 supply-chain agents. AFR-GCN-DDPG differs by learning a bounded con-
505 tinuous correction around a fixed look-ahead anchor, filtering correction tar-
506 gets through common-random-number look-ahead, and controlling patient-
507 condition, capacity, fungible-resource, and patient-indexed specimen-routing
508 decisions in an identity-bound PRM network rather than inventory replen-
509 ishment alone. These studies collectively suggest that graph-based learning
510 is useful for networked supply chain systems; however, most existing work
511 remains focused on prediction, inventory-only control, or general supply-
512 chain planning rather than continuous-action control of PRM-specific capac-
513 ity, reagent, and specimen flows.

514 *3.6. Method foundations: continuous-control reinforcement learning and graph*
515 *neural networks*

516 The actor proposes real-valued pressures for capacity relocation, reagent
517 replenishment and transfer, and specimen routing; a deterministic executor
518 converts those pressures into feasible mixed continuous and integer decisions.
519 This points to continuous-control actor-critic methods coupled with explicit
520 projection. Deterministic policy gradients and DDPG [8] extended actor-
521 critic learning to continuous action spaces, but are known to suffer from
522 value overestimation and training instability. TD3 [9] mitigates overestima-
523 tion through clipped double- Q learning, delayed policy updates, and target
524 smoothing; SAC [10] adds entropy regularization for more robust exploration
525 and stability; and PPO [11] is a widely used on-policy alternative built on a
526 clipped surrogate objective. The practical implication is that DDPG, though
527 historically the default for continuous supply-chain control, is no longer an
528 uncontested choice. A defensible study should compare backbones rather
529 than commit to one, and should expect the more stable off-policy methods
530 to reduce the run-to-run variance that plagues DDPG.

531 Graph neural networks encode relational structure through neighborhood
532 aggregation. The graph convolutional network [12] and its successors (Graph-
533 SAGE [38] for inductive neighborhood sampling and graph attention net-
534 works [39] for learned edge weighting) provide a permutation-equivariant
535 inductive bias that a flat multilayer perceptron lacks. Coupling such en-
536 coders with reinforcement learning has become a recognized paradigm for
537 networked decision problems [17]. For a clinic network whose facilities in-
538 teract through patient-indexed specimen logistics, capacity relocation, and

consumable transshipment, this relational bias is a modeling choice of substance rather than convenience, because it allows a single policy to reason over the topology along which patients, shortages, slack, and disruptions propagate.

3.7. Evaluation considerations for deep reinforcement learning

Evaluation rigor is especially important for DRL-based engineering applications. Henderson et al. [22] showed that deep reinforcement learning results can be sensitive to random seeds, hyperparameters, and implementation details. Agarwal et al. [23] further argued that few-run comparisons can be statistically unreliable and recommended more robust aggregate metrics and interval estimates. These findings imply that a DRL-based supply chain study should report implementation details, training budgets, out-of-sample test procedures, and variability across replications or random seeds when possible. In the present study, we therefore emphasize common simulation settings across all benchmark policies, detailed implementation parameters for the graph-aware actor-critic agents, and Monte Carlo evaluation under multiple disruption and demand forecast scenarios, with interquartile-mean performance and confidence intervals reported across random seeds.

3.8. Research gap and design implications

Read together, these literatures imply five design choices for a distributed PRM capacity-planning controller. First, because the decisions are sequential and real-valued, the controller should use continuous-control actor-critic reinforcement learning rather than value-based or discretized methods. Second, because a facility’s value depends on its network position and resources

Table 2: Positioning of this study relative to closely related literature.

Literature stream	Representative studies	Main limitation relative to this work
PRM operations	Simulation and decentralized planning [1, 6, 7]	No learned graph policy combining patient condition, perishability, identity, and coordinated continuous control.
Supply-chain RL	Inventory and constrained control [27, 19, 21, 20]	Primarily inventory or replenishment, not coupled PRM capacity and fungible-resource sharing.
Decentralized MARL	Cooperative inventory control [18, 37]	Not tailored to identity-bound, patient-specific make-to-order manufacturing.
Supply-chain GNNs	Graph forecasting and graph-aware planning [14, 15, 16, 13]	Mostly predictive or inventory-focused rather than continuous PRM network control.
Structured residual RL	Heuristic-guided and baseline-relative learning [30, 34, 35, 36]	No graph-structured PRM application with patient-condition risk and identity constraints.

563 propagate along inter-facility links, its state should be encoded with a graph
 564 rather than a flat vector. Third, because strong operational rules already
 565 solve much of the nominal problem, the learned policy should target bounded,
 566 evidence-supported corrections to a transparent anchor rather than relearn
 567 the complete policy from scratch. Fourth, DDPG provides a natural canon-
 568 ical formulation and continuity with earlier PRM capacity-planning work,
 569 but its known instability requires controlled comparisons with TD3 and sec-
 570 ondary SAC/PPO variants rather than an untested assumption that DDPG
 571 is always best. Fifth, because reinforcement learning results are sensitive
 572 to seeds and implementation details, evaluation should report multi-seed in-
 573 terquartile means with confidence intervals and, given the clinical stakes,
 574 patient-service outcomes such as eligibility and waiting.

575 None of these streams, however, addresses the problem that defines our
 576 setting. PRM operations research motivates adaptive, networked capacity
 577 planning but does not learn relational policies. Supply-chain graph-and-
 578 reinforcement-learning research supplies the relational-control machinery but
 579 treats demand as fungible and abstract, targeting forecasting or inventory-
 580 only control. Cell-therapy reinforcement learning operates on a single biopro-
 581 cess rather than a clinic network. As a result, no prior work models the three
 582 defining properties of PRM capacity planning (a perishable product with a
 583 hard viability deadline, an identity-bound product that cannot be pooled
 584 or substituted across patients, and demand driven by patient deterioration
 585 while waiting) within a single graph-aware continuous-control formulation.
 586 Our contribution closes this gap. We cast the problem as a constrained
 587 graph MDP, build a reusable simulation testbed that realizes it, and study a

588 family of graph-aware actor–critic controllers, pairing a GCN state encoder
589 with DDPG, TD3, SAC, and PPO, against strong flat-state and heuristic
590 baselines.

591 4. Graph-Aware Deep Reinforcement Learning Framework

592 *Overview: a shared graph encoder with interchangeable actor–critic backbones*

593 Distributed manufacturing networks for personalized regenerative medicine
594 (PRM) comprise complex, interconnected facilities whose operating states
595 are best represented as graphs rather than flat vectors. Classical reinforce-
596 ment learning that treats each facility as independent cannot capture these
597 spatial dependencies. We therefore adopt a two-part design shared by ev-
598 ery method we study. A graph convolutional network (GCN) encoder maps
599 the networked state to permutation-equivariant, neighborhood-aware node
600 embeddings, composed with a continuous-control actor–critic backbone that
601 maps those embeddings to feasibility-projected capacity, reagent, and re-
602 plenishment actions. Writing the controller as *encoder* $\circ$ *backbone* lets us
603 hold the relational representation fixed while swapping the reinforcement-
604 learning backbone, so that any performance difference between backbones is
605 attributable to the learning algorithm rather than to the state representation.

606 *Primary AFR-GCN-DDPG controller and backbone variants*

607 The primary proposed controller is advantage-filtered residual GCN–DDPG,
608 abbreviated AFR-GCN–DDPG. The acronym names the complete controller
609 rather than a new actor–critic backbone: *residual* means that the DDPG
610 actor modifies, rather than replaces, the MDL-2 anchor; *advantage-filtered*

611 means that nonzero teacher corrections are retained only when paired common-
 612 random-number look-ahead estimates a positive improvement over that an-
 613 chor. This filtering is a teacher-data construction and regularization step
 614 inside AFR-GCN-DDPG, not a separate deployed policy. The controller
 615 combines the relational inductive bias of a GCN with the deterministic
 616 continuous-control mechanism used in earlier PRM capacity-planning re-
 617 search, while reducing the burden of relearning a complete operational policy
 618 from scratch. Following residual and constrained-residual control [33, 34], the
 619 actor learns only a bounded correction to a transparent look-ahead anchor.
 620 We also evaluate TD3, SAC, and PPO behind the same GCN encoder (Ta-
 621 ble 4) as controlled backbone variants: TD3 through clipped double- Q learn-
 622 ing, delayed policy updates, and target smoothing; SAC through entropy-
 623 regularized stochastic policies; and PPO through an on-policy clipped surro-
 624 gate objective. All four share the graph representation, feasibility-projection
 625 layer, environment, and evaluation protocol, so backbone comparisons do
 626 not erase AFR-GCN-DDPG’s role as the canonical proposed instantiation.
 627 A GCN encodes relational structure but has no temporal memory of its own;
 628 the actor-critic component supplies the sequential decision dynamics, and
 629 the two together encode spatial correlations and temporal trade-offs jointly.

630 We therefore present the proposed AFR-GCN-DDPG controller in full
 631 (the anchor-plus-residual action, GCN encoder, actor-critic updates, advantage-
 632 filtered supervision, and feasibility projection) and then note where each
 633 stability-oriented variant modifies the critic target, policy parameterization,
 634 or update schedule; the graph encoder and feasibility-projection layer are
 635 identical across backbones.

Table 3: Terminology for the proposed controller and its matched comparisons.

Term	Meaning and role in this study
GCN; DDPG; TD3; SAC; PPO	Established acronyms. Standard graph-encoding and actor-critic algorithms, used with their conventional meanings and cited source formulations.
MDL-2	Study baseline. Two-period mean-demand look-ahead heuristic and transparent anchor; the label is not a general reinforcement-learning acronym.
Advantage-filtered distillation	Training step. Common-random-number look-ahead retains a nonzero teacher correction only when it improves on the anchor. It is not a separate policy.
AFR	Study-specific prefix. Shorthand for the complete <i>advantage-filtered residual</i> design: anchor, bounded learned correction, teacher regularization, feasibility projection, and deployment calibration.
AFR-GCN-DDPG	Proposed controller. AFR with a GCN state encoder and DDPG actor-critic backbone.
AFR-Flat-DDPG; AFR-GCN-TD3	Matched comparisons. The first removes the graph encoder; the second changes the RL backbone. Neither is a separate proposed framework.

Table 4: Continuous-control actor-critic backbones evaluated behind the shared GCN encoder.

Backbone	Policy class	Sampling	Key mechanism and role in this study
DDPG [8]	Deterministic	Off-policy	Canonical backbone of the proposed AFR-GCN-DDPG controller; retains continuity with prior PRM capacity-planning work.
TD3 [9]	Deterministic	Off-policy	Stability-enhanced variant using clipped double- Q learning, delayed policy updates, and target smoothing.
SAC [10]	Stochastic	Off-policy	Entropy-regularized objective for more robust exploration and training stability.
PPO [11]	Stochastic	On-policy	Clipped surrogate objective; an on-policy alternative to the off-policy backbones.

Figure 2 shows the graph abstraction used by the proposed graph-aware framework. Each healthcare clinic is represented as a node with facility-specific state features, including waiting and inbound specimens and their condition summary, reagent inventory, available bioreactor capacity, and other local operational conditions. The edges encode different forms of inter-facility dependency. Information-flow edges let the model propagate operational and patient-condition signals among neighboring clinics; resource-sharing edges represent feasible reagent transshipment relationships; capacity-sharing edges connect each clinic to the central capacity hub; and an explicitly configured subset of qualified inter-clinic links permits identity-preserving routing of intact waiting specimen lots. By applying graph convolution over this structure, the model learns a network-aware representation in which each facility’s decision is informed not only by its own state but also by the states of connected facilities. This graph embedding is then passed to the actor-critic backbone to support coordinated specimen-routing, capacity, reagent, and replenishment decisions.

Formally, let $G_t = (V, E_t)$ represent the manufacturing network at epoch t , where each node $v_i \in V$ denotes a facility with state features x_i^t and edge weights e_{ij}^t indicate logistic or transshipment costs. The GCN encodes local and neighborhood information using a layer-wise propagation rule:

$$h_i^{(l+1)} = \sigma\left(W^{(l)} \sum_{j \in \mathcal{N}(i)} \frac{1}{c_{ij}} h_j^{(l)}\right), \quad (12)$$

where $h_i^{(0)} = x_i^t$, $W^{(l)}$ are trainable weights, $\sigma(\cdot)$ is a ReLU activation, and c_{ij} is a degree-based normalization factor. The resulting embedding $H_t = \{h_i^{(L)}\}_{i=1}^N$ captures the current global state of the network.

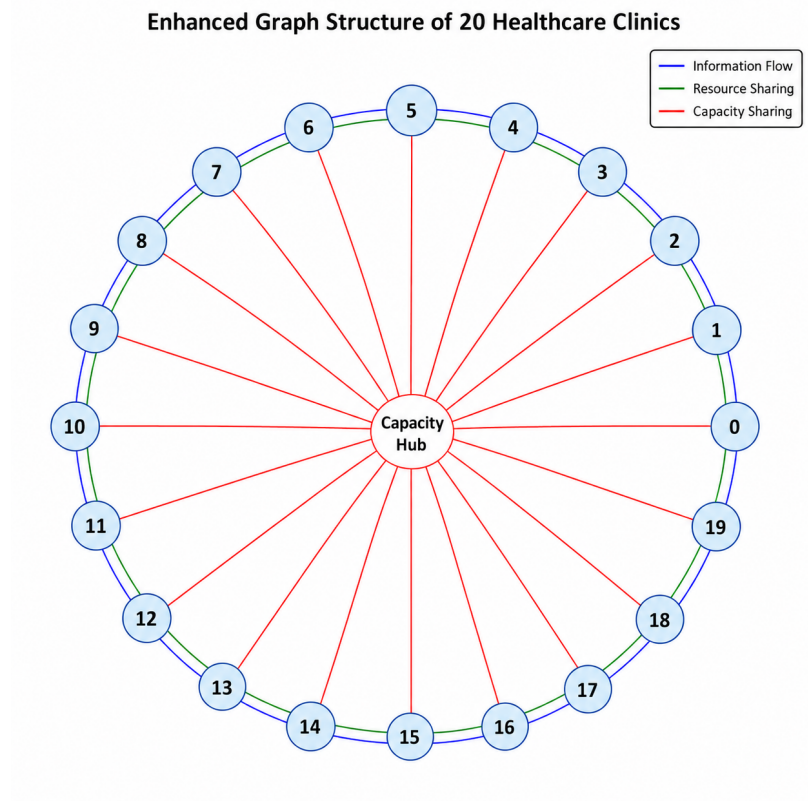


Figure 2: Graph representation of a distributed personalized regenerative medicine manufacturing network with 20 healthcare clinics. Each outer node represents a clinic or manufacturing facility, and the central node represents the shared capacity pool. Blue edges indicate information flow among neighboring clinics, green edges indicate fungible-resource relationships, and red radial edges indicate capacity sharing between each clinic and the central capacity hub. Qualified specimen-logistics links are configured as a constrained subset of inter-clinic relationships and are omitted from this conceptual edge legend. The graph convolutional network encodes local and neighboring state for downstream patient-indexed routing and resource decisions.

659 *Overview of the graph-aware actor-critic workflow (DDPG instantiation)*

660 The method initiates with a dual initialization of the actor and critic
661 networks of the backbone and the GCN for graph state processing. Given
662 an observed state X_t , represented as raw graph data reflecting the current
663 status of the manufacturing network, the GCN is used as a preprocessing
664 unit. It transforms this raw graph into a condensed representation X'_t , which
665 retains the critical structural information of the original graph.

666 Following this transformation, the DDPG algorithm gets involved. Using
667 the graph-processed state X'_t , the actor-network selects an action, which is
668 based on the actor-critic architecture. This action could range from decisions
669 regarding resource allocation to inventory replenishments. Once executed
670 within the manufacturing environment, the system provides a feedback loop
671 in the form of a reward, essentially evaluating the decision's efficacy.

672 The DDPG then uses this feedback, in tandem with the replay buffer,
673 to refine both the actor and critic networks, ensuring better decision-making
674 in subsequent iterations. The GCN, on its part, continuously updates its
675 understanding of the graph structure as the manufacturing network evolves,
676 ensuring it remains adaptive and accurate in its representations.

677 *Initialization*

- 678 • Critic network $Q(X_t, A_t|\theta_Q)$ initialized with weights θ_Q .
- 679 • Actor network $\mu(X_t|\theta_\mu)$ initialized with weights θ_μ .
- 680 • GCN initialized with weights ω for processing graph states.

681 *DDPG Component*

682 The DDPG agent operates on the GCN-encoded state H_t to generate
 683 a bounded residual action $\Delta A_t = \mu(H_t|\theta_\mu)$. Let A_t^{anc} denote the action
 684 produced by the transparent look-ahead anchor and let ρ be a vector of
 685 decision-group-specific trust-region scales. The raw proposed action is

$$A_t^{\text{raw}} = A_t^{\text{anc}} + \rho \odot \Delta A_t. \quad (13)$$

This residual parameterization preserves the anchor’s basic inventory logic while allowing the GCN policy to adapt patient-indexed specimen routing, replenishment, and fungible resource-sharing decisions to network-wide patient pressure, geographic connectivity, disruptions, and demand-prior drift. The critic network $Q(H_t, A_t|\theta_Q)$ estimates the expected cumulative cost. The optimization follows the standard DDPG updates:

$$y_i = c_i(A_i; H_i) + \gamma Q'(H_{i+1}, \mu'(H_{i+1}|\theta_{\mu'})|\theta_{Q'}), \quad (14)$$

$$L_Q = \frac{1}{N} \sum_i (y_i - Q(H_i, A_i|\theta_Q))^2, \quad (15)$$

$$\nabla_{\theta_\mu} J = \frac{1}{N} \sum_i \nabla_a Q(H_i, a|\theta_Q)|_{a=\mu(H_i)} \nabla_{\theta_\mu} \mu(H_i|\theta_\mu). \quad (16)$$

686 *Advantage-filtered residual initialization and deployment*

687 The useful residual corrections in AFR-GCN-DDPG are sparse because
 688 the MDL-2 anchor is already competitive in many states. We therefore ini-
 689 tialize the residual actor through advantage-filtered teacher construction. At
 690 a sampled state, we compare a small set of decision-group residual candi-
 691 dates, including specimen routing and network-resource actions, with the
 692 zero-residual MDL-2 action over a finite look-ahead horizon. Candidate and

693 anchor rollouts use common random numbers so that their score difference is
694 not dominated by different demand, deterioration, or disruption realizations.
695 A nonzero target is retained only when it improves the configured operating-
696 cost and patient-outcome score; otherwise the target is the zero residual. To
697 keep the sparse improvement labels from being overwhelmed, improving and
698 zero-residual examples receive equal total weight. The weighted advantage-
699 filtered data remain in the actor loss during DDPG fine-tuning rather than
700 being used only as a one-time warm start.

701 Deployment uses two independent validation stages. First, checkpoints
702 saved during training are compared on common validation scenarios. Second,
703 the selected checkpoint is evaluated over a fixed set of residual trust-region
704 scales, including zero. The nonzero residual is deployed only when it improves
705 validation cost without reducing service; otherwise the controller falls back to
706 MDL-2. This mechanism keeps the deployed controller within the residual
707 GCN-DDPG family while preventing a poorly generalized correction from
708 overriding the operational anchor.

709 *Reward function*

710 The agents optimize the negative of the immediate cost defined in Sec-
711 tion 2.4. For each epoch t , the reward is

$$r_t = -c(\mathbf{A}_t; \tilde{\mathbf{X}}_t), \quad (17)$$

712 where $c(\cdot)$ is the extended cost of Equation (7), comprising the operational
713 base cost c_{op} (reagent purchase, holding, and shortage; bioreactor holding and
714 shortage; specimen and reagent logistics; and capacity relocation) together

715 with the patient-loss, material-and-product-expiry, and urgency terms. Op-
 716 timizing this reward therefore requires the agent to trade current operating
 717 and logistics cost against the future risk of patient losses and expiry, rather
 718 than resource utilization alone.

719 *Action decoding and feasibility projection..* Because the actor network out-
 720 puts continuous residuals, an additional deterministic decoding and feasibility-
 721 projection step is used before actions are executed in the simulation environ-
 722 ment. The raw anchor-plus-residual action A_t^{raw} is first scaled to the admis-
 723 sible range of each decision variable and then clipped to its variable-specific
 724 lower and upper bounds. The specimen block expresses signed facility pres-
 725 sure rather than anonymous divisible flow. It is scaled by the maximum route
 726 quantity and rounded half away from zero to integer requests; a shared ex-
 727 ecutor then matches opposite requests only across qualified edges and selects
 728 specific eligible patient identities. Reagent transshipment, bioreactor relo-
 729 cation, and replenishment are decoded through their corresponding bounds
 730 and feasibility rules.

731 The decoded action is subsequently passed through a repair operator that
 732 enforces the operational constraints of the manufacturing network. First,
 733 nonnegativity constraints are imposed so that the resulting action cannot
 734 remove more reagents, idle bioreactors, or route-eligible waiting specimen
 735 lots from a facility than are available at that epoch. Second, replenishment
 736 quantities are capped by the maximum purchasable reagent quantity P_t^n and
 737 by supplier availability. Third, network conservation is enforced separately
 738 for fungible resources and integer specimen lots. A specimen route is legal
 739 only for a waiting identity that has not previously moved; each executed

Algorithm 1 Graph-aware actor–critic for distributed PRM capacity planning (DDPG instantiation; TD3/SAC/PPO swap the critic target, policy parameterization, or update schedule).

Require: Anchor policy π_{anc} , residual candidates $\mathcal{D}$, look-ahead horizon L , discount factor γ , soft-update rate τ , actor and critic learning rates α_μ, α_Q , exploration scale σ , minibatch size N .

- 1: Initialize GCN encoder g_ω , actor $\mu(\cdot \mid \theta_\mu)$, critic $Q(\cdot, \cdot \mid \theta_Q)$, target networks μ' and Q' , and replay buffer $\mathcal{R}$.
 - 2: Build weighted teacher data by comparing $\pi_{\text{anc}}(X) + \delta$, $\delta \in \mathcal{D}$, with $\pi_{\text{anc}}(X)$ under common-random-number L -step rollouts; use target δ^* only when it improves the score and zero otherwise.
 - 3: Fit the residual actor to the weighted advantage-filtered targets and retain this loss as an actor regularizer.
 - 4: **for** each environment step t **do**
 - 5: Observe raw graph state X_t and compute graph embedding $Z_t = g_\omega(X_t)$.
 - 6: Select exploratory residual action $A_t = \pi_{\text{anc}}(X_t) + \rho \odot \mu(Z_t \mid \theta_\mu) + \epsilon_t$, where $\epsilon_t \sim \mathcal{N}(0, \sigma^2 I)$.
 - 7: Project or clip A_t to the feasible set and execute the resulting action $\hat{A}_t$.
 - 8: Observe cost/reward r_t , next state X_{t+1} , and store $(X_t, \hat{A}_t, r_t, X_{t+1})$ in $\mathcal{R}$.
 - 9: **if** an update is scheduled **then**
 - 10: Sample minibatch $(X_i, A_i, r_i, X_{i+1})_{i=1}^N$ from $\mathcal{R}$ and compute $Z_i = g_\omega(X_i)$, $Z_{i+1} = g_\omega(X_{i+1})$.
 - 11: Compute targets $y_i = r_i + \gamma Q'(Z_{i+1}, \mu'(Z_{i+1} \mid \theta_{\mu'}) \mid \theta_{Q'})$.
 - 12: Update critic by minimizing $L_Q = N^{-1} \sum_i (y_i - Q(Z_i, A_i \mid \theta_Q))^2$.
 - 13: Update actor using the deterministic policy gradient plus the weighted advantage-filtered regularizer.
 - 14: Backpropagate through the GCN encoder and softly update target networks.
 - 15: **end if**
 - 16: **end for**
 - 17: Select checkpoint and residual scale on independent validation scenarios; return the calibrated residual policy or the MDL-2 fallback.
-

740 donor event has one matched receiver event, and unmatched requests are
 741 recorded as blocked rather than repaired by substitution. Finally, all transfer
 742 and routing actions are restricted to their configured graph edges. Finished-
 743 product return is a deterministic identity-preserving lifecycle transition, not
 744 an actor decision.

745 Equivalently, the executed action can be interpreted as the feasible action
 746 $\hat{A}_t$ obtained by projecting the decoded actor output onto the feasible action
 747 set $\mathcal{F}(X_t)$:

$$\hat{A}_t = \Pi_{\mathcal{F}(X_t)}(A_t^{\text{raw}}), \quad (18)$$

748 where $\mathcal{F}(X_t)$ is determined by current facility states, patient lifecycle and
 749 route history, inventory availability, capacity availability, supplier limits, the
 750 autologous identity constraint, and graph connectivity. This projection step
 751 improves engineering credibility by ensuring that all actions executed in the
 752 simulation correspond to implementable specimen-routing, capacity, replen-
 753 ishment, and transshipment decisions.

754 Each stability-oriented backbone reuses this encoder and projection layer
 755 unchanged and alters only the reinforcement-learning update. TD3 replaces
 756 the single critic with clipped double- Q targets, delays actor updates, and
 757 adds target-policy smoothing; SAC replaces the deterministic actor with an
 758 entropy-regularized stochastic policy and a soft value target; and PPO re-
 759 places off-policy replay with on-policy rollouts optimized under a clipped
 760 surrogate objective. In all cases the actor and critic operate on the GCN
 761 embedding Z_t rather than on the flat state, and the executed action is the
 762 feasibility-projected $\hat{A}_t$.

763 *Summary*

764 The proposed method is therefore a residual GCN-DDPG controller: a
765 GCN encoder supplies a permutation-equivariant, network-aware state repre-
766 sentation; a deterministic actor learns bounded corrections to a transparent
767 look-ahead anchor; and a feasibility-projection layer guarantees that every
768 executed action respects inventory, capacity, supplier, patient identity, route
769 history, and connectivity constraints. The routing-primary architecture com-
770 parison holds specimen-routing availability, the anchor, residual action space,
771 advantage-filtered teacher protocol, projection, training budget, and evalu-
772 ation seeds fixed between GCN and parameter-matched flat encoders. The
773 earlier backbone comparison instead holds the graph encoder fixed while
774 replacing DDPG with TD3. Residual SAC and PPO are secondary fam-
775 ily extensions because their stochastic residual distributions and likelihood
776 calculations require additional matched implementation work. These vari-
777 ants test backbone robustness without redefining them as separate proposed
778 methods.

779 **5. Computational Experiments**

780 This section reports three empirical stages designed to separate anchor
781 improvement, graph representation, and online reinforcement-learning attri-
782 bution. The first stage calibrates patient-condition dynamics and evaluates
783 replenishment residuals under demand-prior drift. The second expands the
784 action space to geographically structured reagent and idle-capacity transfers
785 and compares the graph policy with a parameter-matched flat policy. The
786 third holds the pretrained graph controller fixed while testing whether conser-

787 vative online TD3 updates add value beyond frozen pretraining. The exper-
788 iments compare the learned graph-aware correction policy with its heuristic
789 anchor, other heuristic and look-ahead policies, flat-state residual policies,
790 and graph-aware backbone variants under the same simulation environment.

791 5.1. Objective

792 The objective of this study is to compare the proposed residual GCN-
793 DDPG framework with traditional heuristic methods, its MDL-2 anchor, flat-
794 state residual DDPG, and stability-oriented graph-aware actor-critic variants
795 in distributed manufacturing.

796 5.2. Methodology

797 5.2.1. Benchmarking Setup

798 The final benchmark is organized as a hierarchy of matched comparisons
799 rather than a single undifferentiated algorithm ranking (Table 5). The first
800 comparison tests whether reinforcement learning improves the MDL-2 an-
801 chor. The second isolates the graph representation by replacing the GCN
802 with a flat MLP while keeping the entire residual-learning recipe fixed. The
803 third tests whether the result is specific to DDPG by repeating the residual
804 comparison with TD3. Online rollout shields and the supervised GCN shield
805 selector are reported separately because the former uses additional simula-
806 tion at decision time and the latter is not an RL policy. Pure end-to-end RL
807 and the stochastic SAC/PPO family are reported as secondary comparators
808 rather than being allowed to substitute for these mandatory tests.

809 The heuristic baselines are:

- 810 • **MYO (Myopic):** This heuristic minimizes single-period cost without
811 considering future demand.
- 812 • **MDL-1 (One-period Mean Demand Lookahead):** MDL-1 ex-
813 tends the planning horizon to one lookahead period using deterministic
814 state transitions based on mean demand.
- 815 • **MDL-2 (Two-period Mean Demand Lookahead):** MDL-2 ex-
816 tends the planning horizon to two lookahead periods using determinis-
817 tic state transitions based on mean demand.
- 818 • **ISO (Isolated Regional Facilities):** ISO assumes no resource shar-
819 ing between facilities and focuses only on local replenishment decisions.
- 820 • **F-MYO (Forecast Myopic):** F-MYO augments current-period net-
821 work balancing with the demand forecast available at the decision
822 epoch.
- 823 • **uMYO and pMYO (Patient-aware myopic policies):** uMYO
824 adds urgency-based replenishment and capacity surges, while pMYO
825 uses a bounded patient-risk workload uplift. These controls prevent
826 the proposed method from being compared only with condition-blind
827 heuristics.

828 The learning-based comparisons are:

- 829 • **MLP residual DDPG / flat-state residual DDPG:** This base-
830 line uses the same MDL-2 anchor, residual action space, DDPG actor-
831 critic structure, training budget, and feasibility projection as the pro-

posed method, but removes the GCN encoder. Instead, all facility-level states are concatenated into a flat vector and passed directly into multilayer perceptron actor and critic networks. This baseline is designed to answer whether performance improvements come from reinforcement learning alone or from explicitly encoding graph-structured inter-facility dependencies.

- **Residual GCN–TD3 and matched flat residual TD3:** These policies keep the anchor, advantage-filtered supervision, action constraints, training data, and evaluation streams fixed while replacing the DDPG update with TD3. They test whether twin critics, delayed actor updates, and target smoothing improve stability.
- **Pure GCN and flat actor–critic policies:** From-scratch DDPG and TD3 quantify the benefit of anchoring; SAC and PPO provide secondary off-policy stochastic and on-policy comparisons. Residual SAC/PPO are included only after their residual-space probability calculations and matched flat variants pass the same validation protocol.

In addition, graph ablation experiments are planned to evaluate the role of different edge types. Specifically, we consider removing capacity-sharing edges and removing resource-sharing edges to assess whether these relational structures contribute to policy performance under disruption and demand uncertainty.

All methods receive the same decision-time observations and are evaluated on identical paired random-number streams. In demand-prior-drift experiments, heuristics receive the same misspecified demand prior available

Table 5: Final comparison hierarchy for the proposed controller.

Comparison layer	Required policies	Question answered
Anchor improvement	im- AFR-GCN-DDPG versus MDL-2	Does the learned correction improve the strong policy from which it starts?
Strong heuristics	heuristic- ISO, MYO, F-MYO, uMYO, pMYO, MDL-1, and MDL-2	Is the controller competitive with sharing, forecast-aware, patient-aware, and look-ahead rules?
Representation ablation	AFR-GCN-DDPG versus matched AFR-flat-DDPG	Is any gain attributable to graph representation rather than residual learning alone?
Backbone robustness	ro- Residual GCN-DDPG versus residual GCN-TD3, each with a matched flat counterpart	Does the result persist when the deterministic actor-critic backbone changes?
Anchor ablation	abla- Residual versus pure GCN/flat DDPG and TD3	Does the heuristic anchor improve sample efficiency, stability, and final performance?
Deployment reference	MDL-2/pMYO rollout shields and supervised GCN shield selector	How does policy quality trade off against online simulation cost, and can RL outperform non-RL distillation?
Secondary family	GCN/flat SAC and PPO; residual forms after matched implementation	Is the conclusion robust to stochastic off-policy and on-policy learning?
Graph mechanism	Full graph, no/shuffled edges and geographic-edge ablations	Does the policy use topology rather than merely additional parameters?

856 to the controller and no method receives the realized future demand rates.
857 Learned policies use the same training-scenario distribution, environment-
858 interaction budget, checkpoint-selection budget, and held-out evaluation seeds
859 within each matched comparison. Online decision time and the number of
860 simulator rollouts are reported separately for rollout-based methods. This
861 prevents a performance difference from being created by unequal forecasts,
862 simulator access, evaluation noise, or hidden online computation.

863 5.2.2. *Performance Metrics*

864 We evaluate each method using both objective-based and engineering-
865 relevant performance metrics. The primary metric is the average total weighted
866 objective J_{tot} in Equation (8). It includes the complete operating-cost group
867 (reagent purchase, holding, and shortage; bioreactor holding and shortage;
868 and reagent and capacity transfer) and the patient-related penalty group
869 (patient loss, material or therapy expiry, and urgency). The percentage im-
870 provement reported in headline comparisons always uses the anchor’s com-
871 plete J_{tot} as the denominator, as in Equation (9); we do not remove large
872 components after observing the result.

873 Because a weighted total can be dominated by patient-loss or capacity-
874 shortage penalties, every formal comparison also reports the absolute paired
875 difference and an additive cost decomposition. Component-specific percent-
876 ages are secondary mechanism measures: they reveal whether a policy saves
877 purchase, holding, shortage, transfer, or patient-penalty cost, but do not
878 replace the primary total-objective comparison. When an oracle or other
879 defensible lower-cost reference is evaluated, we also report the headroom-
880 capture measure in Equation (10). To strengthen the application relevance

881 of the experiments, we track the following secondary metrics:

- 882 • **Cost decomposition:** operating/base cost and its reagent, bioreac-
883 tor, and transfer components, together with patient-loss, expiry, and
884 urgency penalties.
- 885 • **Completion service level:** the fraction of eligible patient demand
886 completed within the planning horizon.
- 887 • **Patient loss and manufacturing-period ineligibility:** the number
888 of patients who become ineligible or expire before treatment and the
889 fraction who become ineligible after manufacturing begins.
- 890 • **Average waiting time / fulfillment delay:** the average number of
891 epochs that patient specimens wait before manufacturing begins.
- 892 • **Reagent shortage frequency:** the frequency with which manufac-
893 turing is delayed because insufficient reagents are available.
- 894 • **Bioreactor utilization:** the fraction of available bioreactor capacity
895 used for production across facilities.
- 896 • **Routing, transshipment, and relocation activity:** patient-indexed
897 specimen route count, distance, transit time, route cost, and blocked
898 requests, together with reagent movements and idle-bioreactor relo-
899 cations. These are mechanism diagnostics rather than evidence that
900 routing is itself a contribution.
- 901 • **Inference time:** the computational time required for a trained policy
902 or heuristic to generate one epoch-level decision.

903 These metrics are included because the objective of PRM capacity plan-
 904 ning is not only to reduce a modeled weighted objective, but also to support
 905 timely patient-specific manufacturing, effective resource utilization, and reli-
 906 able system performance under demand and supply uncertainty. Cost-weight
 907 sensitivity is therefore required before translating an objective difference into
 908 an economic claim.

909 5.2.3. *Heuristic Implementations*

910 We provide brief descriptions of the heuristic methods implemented:

- 911 • **MYO (Myopic):** MYO focuses on minimizing costs within the cur-
 912 rent period without considering future consequences.
- 913 • **MDL-1 and MDL-2:** These methods extend the planning horizon to
 914 one and two lookahead periods, respectively, using deterministic state
 915 transitions based on mean demand.
- 916 • **ISO (Isolated Regional Facilities):** ISO assumes no sharing of re-
 917 sources between manufacturing facilities and primarily concentrates on
 918 reagent replenishment.

919 5.2.4. *Implementation and statistical protocol*

920 The models are implemented in Python using PyTorch, and all experi-
 921 ment parameters, training seeds, checkpoint candidates, deployment scales,
 922 and evaluation streams are stored in configuration files. Within a matched
 923 comparison, the graph and flat policies receive the same state variables, an-
 924 chor actions, teacher targets, environment interactions, validation budget,
 925 and common-random-number (CRN) holdout episodes. The graph policy

926 replaces the flat encoder with message passing over the clinic graph; the re-
927 maining actor-critic and deployment components are held fixed or parameter
928 matched.

929 *Routing-primary hyperparameter development.* The routing-primary DDPG
930 settings were obtained through sequential, development-only sensitivity screens
931 rather than an exhaustive automated HPO search. The screens first bounded
932 the residual correction and then varied the online update schedule, actor
933 learning rate, exploration noise, imitation and frozen-reference regularization,
934 and deployment gate. Mechanism-development screens separately examined
935 integer-lot straight-through specimen actions, anchor-relative rewards, re-
936 turn horizon, and evidence required to release the frozen-pretraining refer-
937 ence. Table 6 summarizes the principal ranges and the current candidate.
938 All screens used development seeds and CRN streams; none is included in a
939 confirmatory estimate.

940 The selected candidate uses batch size 64, critic learning rate 3×10^{-4} ,
941 integer specimen-lot execution with a straight-through actor gradient, and
942 specimen-only residual corrections; reagent, capacity, and replenishment resid-
943 ual groups remain zero in this screen. The most informative warning against
944 single-seed selection came from the update-frequency study: the frequency-4
945 candidate passed its seed-0 screen but did not generalize in the subsequent
946 three-seed confirmation. Seed 0 and the earlier routing seeds 1–2 are there-
947 fore treated as development evidence, not untouched confirmation seeds.

948 The replenishment-residual DDPG and TD3 confirmation uses 300 train-
949 ing episodes for each of five seeds, checkpoint candidates between episodes
950 50 and 300, and 100 independent paired holdout replications per seed. Re-

Table 6: Routing-primary DDPG development screens. The selected values define the current candidate but are not themselves confirmatory results.

Component	Development values or variants	Selected value
Residual deployment magnitude	Anchor-only; 0.10, 0.25, and 0.50	0.10 for specimen transfer
Environment update frequency	Every 1 or 4 simulator steps	1
Actor update frequency	Every 2, 4, or 8 critic updates	2
Online actor learning rate	1×10^{-5} or 3×10^{-5}	1×10^{-5}
OU exploration standard deviation	0.005, 0.02, or 0.05	0.005
Imitation/reference regularization	Imitation 1 or 2; reference 0, 250, or 500	1 and 500, respectively
Pretraining/critic warm start	Short smoke or 300 epochs; 0 or 500 critic updates	300 epochs and 500 updates
Online return evidence	Immediate or four-step anchor-relative return; unfiltered, positive, or dual confirmation	Four-step dual confirmation
Material improvement threshold	Zero or one patient-loss-equivalent scaled cost	0.0005
Deployment gate	Fixed development validation grid with anchor fallback	0.575

ported confidence intervals use a two-level bootstrap over training seeds and paired evaluation replications. The regional network-action experiment uses three independently initialized graph and flat policies, 30 paired validation episodes, and 100 previously unseen paired holdout replications per seed. Because its selected checkpoint is dominated by advantage-filtered teacher distillation, that experiment is interpreted as graph-policy evidence rather than as independent evidence for online DDPG updates.

The conservative online-TD3 screen starts from the corresponding frozen pretrained graph and flat checkpoints and runs 100 online episodes for seeds 0, 1, and 2. Each run completed 5,200 update calls, including 5,073 critic updates and 2,286 delayed actor updates. Final and frozen checkpoints are evaluated with identical CRN keys, using 20 validation and 100 holdout replications per seed for the final policies and 5 validation and the same 100 holdout replications per seed for the frozen controls. This campaign was run on an NVIDIA RTX 4090 GPU; all optimization diagnostics were finite, and no CPU fallback occurred. These experiment-specific budgets replace the earlier generic one-million-step description and make the reported results traceable to the executed configurations.

The routing-primary campaign is prospectively separated from those earlier no-routing results. It holds patient-indexed specimen routing enabled for every primary policy and compares GCN residual DDPG, a parameter-matched flat residual DDPG controller, and the MDL-2 routing anchor. One fresh routing-enabled teacher is shared across the learned architectures. After the development screens are frozen, confirmatory training uses five untouched seeds (10–14) for each learned architecture. Every run first re-

ceives 100 episodes, or approximately 5,200 online transitions, with complete state saved every five episodes. Prespecified development learning curves at episodes 25, 50, 75, and 100 determine one global budget decision before the final holdout is opened: if either architecture has not plateaued, all ten runs are extended to 200 episodes. Extension to the common maximum of 300 is allowed only if the 100–200 curves show continuing multi-seed improvement without clinical-guardrail deterioration. A method- or seed-specific extension is not permitted.

Final and frozen-pretraining checkpoints are evaluated on four routing scenarios with at least 100 paired CRN replications per training seed; additional evaluation replication may narrow uncertainty without changing the trained policies. Lead-zero specimen transit and one-epoch product return are prespecified sensitivities. The primary attribution is GCN versus matched flat, GCN versus MDL-2, and final versus frozen pretraining. No-routing is retained only for deterministic mechanics regression or a separately approved supplementary ablation, not as a default formal training arm. Development screens are reported as tuning evidence only, and no routing-primary headline claim is made until the untouched-seed outputs pass the locked gates.

5.3. *distributed manufacturing simulation*

To evaluate the proposed method, we now construct a simulation environment of a decentralized manufacturing system consisting of $N = 20$ manufacturing facilities and its dedicated reagent supplier. Each PRM product requires $T = 3$ epochs of manufacturing time, and the considered finite problem horizon contains $\Gamma = 52$ decision epochs. For every epoch t , there

Algorithm 2 State normalization

Require: State variables: s_t^n , r_t^n , b_t^n , and p_t^n

1: Maximal values of the state variables for all facilities: M_s , M_r , M_b , and M_p

Ensure: Normalized state variables: $\hat{s}_t^n$, $\hat{r}_t^n$, $\hat{b}_t^n$, $\hat{p}_t^n$

2: **for** $n = 1$ to N **do**

3: $\hat{s}_t^n = s_t^n / M_s$

4: $\hat{r}_t^n = r_t^n / M_r$

5: $\hat{b}_t^n = b_t^n / M_b$

6: $\hat{p}_t^n = p_t^n / M_p$

7: **end for**

1001 will be d_t^n new demand for facility n . The demand arrival process for facility
1002 n follows a Poisson distribution $\text{Poisson}(\lambda_n)$. The supplier disruption follows
1003 independent Bernoulli processes $\text{Ber}(\zeta_n)$, i.e., the facility n has a probability
1004 ζ_n to have supplier disruption and could not purchase any reagent for the
1005 epoch. This simulation can be further adjusted to reflect any real situation,
1006 e.g., supply disruption and demand uncertainty, that may be encountered.
1007 Such flexibility in using the simulation system will be demonstrated in Sec-
1008 tion 5.4 when we compare GCN-DDPG with other methods.

1009 Patient-condition parameters are calibrated before policy comparison.
1010 The nominal scenario uses $\lambda_H = 0.0013$, $\lambda_F = 0.0156$, Weibull shape $\kappa = 1.5$
1011 and scale $\theta = 4$, post-deterioration acceleration $\rho = 3$, eligibility threshold
1012 $\epsilon = 0.80$, and waiting-time pressure $0.0013\alpha^2$. We selected these values from
1013 a fixed-seed sweep whose pre-specified target was a 10–15% *patient ineligi-*
1014 *bility during manufacturing* rate over the three-week process. This outcome

1015 is not labeled a technical manufacturing failure: it records a patient who
 1016 becomes clinically ineligible after production begins, whereas a process or
 1017 release-specification failure would be a separate stochastic event. The target
 1018 is consistent with the order of magnitude of pre-infusion attrition reported for
 1019 autologous CAR-T pathways [3] and with a 5% weekly eligibility-transition
 1020 benchmark, which gives $1 - (1 - 0.05)^3 = 14.26\%$ over three weeks.

1021 The patient-facing weights are $\omega_{\text{lost}} = 500,000$, $\omega_{\text{exp}} = 100,000$, and $\omega_{\text{urg}} =$
 1022 $25,000$. The earlier exploratory values (50,000, 40,000, 5,000) made one-time
 1023 patient loss cheaper than even one period of combined reagent and bioreactor
 1024 shortage, so a policy could lower its objective when a patient left the system.
 1025 The calibrated ordering removes this directional inconsistency and is held
 1026 fixed for all policy comparisons.

1027 The epoch is assumed to occur weekly. Reagent and idle-bioreactor trans-
 1028 fers enter an in-transit pipeline and arrive after one, two, or three epochs,
 1029 according to whether the geographic distance is at most 500 miles, between
 1030 500 and 1,500 miles, or above 1,500 miles. In the routing-primary campaign,
 1031 an intact waiting specimen may take one direct qualified route. The main
 1032 setting uses one transit epoch and immediate modeled return of finished
 1033 product to the collection facility; zero-epoch specimen transit and one-epoch
 1034 product return are sensitivities. The previously reported case studies set
 1035 $w_t^n = 0$ and are therefore not routing-effect experiments. After demand
 1036 and any due transfers or routes have arrived, the next observed state is
 1037 $\mathbf{x}_{t+1}^n = (s_{t+1}^n, r_{t+1}^n, \mathbf{b}^n, P_{t+1})$ for facility $n = 1, \dots, N$, according to Equation
 1038 (1). The state variables are updated by the following rules:

$$\begin{aligned}
s_{t+1}^n &= s_t^n - m_t^n + d_t^n + w_t^n \\
r_{t+1}^n &= r_t^n - m_t^n + e_t^n + p_t^n \\
b_{t+1}^{n,\tau} &= \begin{cases} b_t^{n,0} - m_t^n + b_t^{n,1} + q_t^n & \text{if } \tau = 0 \\ b_t^{n,\tau+1} & \text{if } 1 \leq \tau \leq T-2 \\ m_t^n & \text{if } \tau = T-1 \end{cases}
\end{aligned}$$

1039 where $m_t^n = \min(s_t^n, b_t^{n,0}, r_t^n)$ is the number of products that begin man-
1040 ufacturing at epoch t , d_t^n is the demand arriving at facility n in epoch t , and
1041 P_{t+1} follows an exogenous Markov chain from P_t . The operating component
1042 of the single-period objective can be decomposed as

$$\begin{aligned}
c_{\text{op}}(\mathbf{A}_t; \mathbf{X}_t) &= \sum_{n=1}^N C_{\text{op},t}^n, \\
C_{\text{op},t}^n &= C_{\text{Reg},t}^n + C_{\text{Bio},t}^n + C_{\text{Tran},t}^n.
\end{aligned}$$

1043 The meanings of $C_{\text{Reg},t}^n$, $C_{\text{Bio},t}^n$, and $C_{\text{Tran},t}^n$ are summarized below. These
1044 terms constitute only c_{op} ; the patient-related terms in Equation (7) must
1045 also be included to obtain the total weighted objective.

- 1046 • $C_{\text{Reg},t}^n = c_R p_t^n + h_R (r_{t+1}^n - s_{t+1}^n)^+ + p_R (s_{t+1}^n - r_{t+1}^n)^+$ is the cost related
1047 to reagent at epoch t , including the reagent replenishment cost $c_R p_t^n$,
1048 the reagent overstock holding cost $h_R (r_{t+1}^n - s_{t+1}^n)^+$, and the reagent
1049 understock penalty $p_R (s_{t+1}^n - r_{t+1}^n)^+$;
- 1050 • $C_{\text{Bio},t}^n = h_B (b_{t+1}^{n,0} - s_{t+1}^n)^+ + p_B (s_{t+1}^n - b_{t+1}^{n,0})^+$ is the cost related to

1051 bioreactor at epoch t , including the bioreactor overstock holding cost
1052 $h_B (b_{t+1}^{n,0} - s_{t+1}^n)^+$, and the bioreactor understock penalty $p_B (s_{t+1}^n - b_{t+1}^{n,0})^+$;
1053 • $C_{\text{Tran},t}^n = K_R |e_t^n| + K_B |q_t^n| + K_S |w_t^n|$ is the logistics-cost expression.
1054 The routing-primary campaign includes the patient-indexed specimen
1055 term; it vanishes only in the earlier no-routing case studies.

Table 7: Modeled cost coefficients used in the calibrated experiments. Values are weighted-objective units and are held fixed across policies. Transfer coefficients are base values before the geographic distance and travel-time surcharge. Patient-related coefficients are penalty or shadow-value parameters rather than observed cash expenditures.

Component	Coefficient	Interpretation
Reagent purchase, c_R	42,174.0	per purchased unit
Reagent holding, h_R	113.5	per excess unit-period
Reagent shortage, p_R	86,504.5	per shortage unit-period
Bioreactor holding, h_B	14.4	per idle unit-period
Bioreactor shortage, p_B	50,273.6	per shortage unit-period
Reagent transfer, K_R	200.0	per transferred unit, before surcharge
Capacity transfer, K_B	500.0	per relocated unit, before surcharge
Patient loss, ω_{lost}	500,000	per lost patient
Expiry, ω_{exp}	100,000	per wasted material or therapy unit
Urgency, ω_{urg}	25,000	per at-risk unserved patient-period

1056 5.4. Case Study 1: Interconnected Clinic Network Analysis

1057 This case study holds the demand prior equal to the arrival distribution
1058 and varies the supplier disruption rate over 0.05, 0.30, and 0.60. Its purpose

is to measure the value of graph-aware coordination when the deterministic heuristics are not disadvantaged by demand misspecification. The final matched multi-seed campaign for these three regimes is still pending; consequently, we do not report numerical disruption-effect claims in the present draft. This case study will use the same five training seeds, paired Monte Carlo evaluation, checkpoint selection, and two-level bootstrap protocol as Case Study 2.

5.5. Case Study 2: Adaptability to Non-Stationary Demand

This case study evaluates the main robustness hypothesis under patient-condition and geographic network dynamics. The 20 clinics are partitioned into four five-clinic demand groups. MDL-2 plans with Poisson prior rates $[7.5, 3.2, 6.0, 3.5]$, whereas the simulator generates arrivals using true rates $[10.5, 3.52, 7.5, 3.15]$. The environment also includes patient deterioration, expiry, clustered demand shocks, regional supplier disruptions, and distance-dependent transfer lead times. During training, demand rates are randomized between 0.8 and 1.5 of the scenario rates and forecast error between 0 and 0.35; evaluation is fixed to the scenario above.

Before retraining learned policies, we evaluated ISO and MDL-2 with 100 paired Monte Carlo replications to verify that the calibrated patient process and objective produce the intended clinical range. Table 8 reports this calibration pilot and the subsequent matched learned-policy screen. Learned-policy results produced before the complete patient lifecycle and objective calibration are not comparable and are therefore retired rather than carried forward into this experiment.

Both heuristics fall inside the pre-specified clinical calibration band: manufacturing-

Table 8: Calibrated demand-prior-drift results. ISO and MDL-2 use the initial 100-replication calibration stream; learned-policy and matched heuristic rows aggregate 100 paired replications for each of three training seeds. Objective values are the total modeled weighted objective in billions of cost units, not realized accounting expenditure. Lower objective, patient loss, and manufacturing-period ineligibility are better; higher completion service and eligibility are better.

Method	Objective (10^9 units)	Completion	Eligibility	Lost	Mfg. inelig.
Initial ISO calibration	3.990456	0.337612	0.377537	4021.92	0.146398
Initial MDL-2 calibration	3.895694	0.371948	0.421024	3698.87	0.121364
AFR-Flat-DDPG	3.927584	0.370359	0.507030	3730.36	0.122223
AFR-GCN-DDPG	3.927594	0.370359	0.507030	3730.35	0.122223
AFR-GCN-TD3	3.927628	0.370358	0.507030	3730.36	0.122223
MDL-2	3.927620	0.370359	0.507030	3730.36	0.122223
ISO	4.011547	0.337521	0.460044	4043.10	0.146468
pMYO	4.180485	0.343311	0.496776	3894.09	0.133332
MYO	4.194549	0.342889	0.496586	3897.01	0.133582

period patient ineligibility is 14.64% for ISO and 12.14% for MDL-2. MDL-2 reduces the mean total weighted objective by 94.76 million modeled cost units (2.37%) relative to ISO; the paired 95% interval for MDL-2 minus ISO is $[-108.81, -80.71]$ million cost units, and MDL-2 has a lower objective in 92 of 100 paired replications. It also loses 323.05 fewer patients on average, discards 49.40 fewer in-process therapies, and raises completion service by 0.03434.

The calibrated learned-policy screen used 100 training episodes and three training seeds. AFR-GCN-DDPG selected and deployed a nonzero residual in all three seeds; AFR-GCN-TD3 and AFR-Flat-DDPG each returned to MDL-2 in one seed. AFR-GCN-DDPG reduced the mean total weighted objective relative to MDL-2 by only 26,519 modeled cost units (0.000675%),

with a two-level bootstrap 95% interval of $[-89,588, 40,330]$ cost units. Its mean objective was 9,460 cost units higher than the matched flat residual policy, with interval $[-7,951, 24,621]$. Completion service and patient loss were effectively unchanged. Thus the screen demonstrates safeguard-compatible parity with a strong anchor, but it does not support either a material RL improvement or a graph-representation advantage. The accepted correction primarily substitutes additional reagent purchase and holding cost for lower reagent shortage cost. Because this residual changes replenishment only, the next matched experiment expands the bounded correction to distance-aware reagent and idle-bioreactor transfers, where geographic message passing can affect the decision.

The small headline percentage in this screen is intentionally computed against the complete weighted objective rather than a narrower, post hoc denominator. Patient-related penalties and capacity-shortage penalties can dominate that total, while a replenishment-only residual directly changes only a subset of the operating components. We therefore interpret 0.000675% as statistical parity rather than as a material economic saving. Subsequent network-residual comparisons report the total-objective change together with operating, bioreactor-shortage, transfer, and patient-penalty decompositions so that denominator dilution and the mechanism of any improvement remain visible.

5.5.1. Formal five-seed demand-prior-drift confirmation

The formal replenishment-residual comparison increased the training budget to 300 episodes and used five independent training seeds. Each selected checkpoint was evaluated on 100 paired holdout replications, yielding 500

1121 paired outcomes per method. Table 9 reports the mean outcomes, and Ta-
 1122 ble 10 reports paired differences. The learned-policy values in these tables
 1123 supersede the 100-episode diagnostic screen in Table 8.

Table 9: Five-seed demand-prior-drift confirmation. Objective values are the total modeled weighted objective in billions of units. Each row aggregates 100 paired holdout replications for each of five training seeds.

Method	Objective (10^9 units)	Completion	Eligibility	Waiting	Lost	At-risk unserved
MDL-2	1.268257	0.429982	0.530313	3.052625	3665.59	5206.41
AFR-Flat-DDPG	1.268089	0.430320	0.530523	3.052012	3663.28	5203.86
AFR-GCN-DDPG	1.268049	0.430321	0.530524	3.052012	3663.27	5203.86
AFR-GCN-TD3	1.268241	0.430317	0.530523	3.052019	3663.30	5203.88

Table 10: Paired five-seed demand-prior-drift comparisons. Differences are candidate minus baseline, so a negative objective difference favors the first method. Confidence intervals use the two-level bootstrap over training seeds and paired replications.

Comparison	ΔJ_{tot} (units)	Relative gap	Paired 95% CI	Wins / 500
AFR-GCN-DDPG vs. MDL-2	-207,485	-0.016360%	[-332,641, -91,367]	323
AFR-Flat-DDPG vs. MDL-2	-167,610	-0.013216%	[-286,899, -46,230]	319
AFR-GCN-DDPG vs. AFR-Flat-DDPG	-39,876	-0.003145%	[-89,702, 14,025]	285
AFR-GCN-TD3 vs. MDL-2	-15,823	-0.001248%	[-142,607, 99,437]	268
AFR-GCN-DDPG vs. AFR-GCN-TD3	-191,662	-0.015112%	[-234,659, -148,401]	377

1124 AFR-GCN-DDPG improves on MDL-2 in every training-seed mean and
 1125 has a two-level confidence interval fully below zero. Completion service in-
 1126 creases by 0.000340, eligibility by 0.000211, average waiting time falls by
 1127 0.000612, patients lost fall by 2.32, and at-risk unserved patients fall by 2.55.
 1128 This supports a statistically stable anchor improvement under demand-prior
 1129 drift, but not a large economic effect. The matched flat policy also improves
 1130 on MDL-2, and the direct graph-versus-flat interval includes zero. Replen-

1131 ishment residual learning is therefore supported in this scenario, whereas an
1132 independent graph-representation advantage is not.

1133 The matched TD3 policy preserves the patient-facing improvements but
1134 has a cost interval crossing zero. DDPG is lower cost than TD3 in all five
1135 training-seed means, and their paired interval is fully below zero. This
1136 comparison supports DDPG as the canonical backbone for the proposed
1137 replenishment-residual controller; it is not a claim that DDPG generally
1138 dominates TD3 outside this residual-learning design.

1139 *5.6. Case Study 3: Graph Attribution Under Regional Non-Stationarity*

1140 This earlier no-routing case study combines patient-condition dynamics,
1141 clinic geography, distance-based transfer delays, regional demand heterogene-
1142 ity, and demand regime changes. Unlike the replenishment-only experiment,
1143 the network residual contains a node head for signed replenishment correc-
1144 tions and edge heads for reagent and idle-bioreactor transfer corrections.
1145 Edge flows are projected to preserve resource conservation and feasibility,
1146 while patient-indexed specimen routing is disabled by that historical proto-
1147 col. This action space gives graph message passing a direct route to affect
1148 geographically structured decisions.

1149 *5.6.1. Network AFR-GCN-DDPG under an abrupt regional shift*

1150 The abrupt-shift policy uses transfer-focused advantage-filtered targets
1151 and a fixed deployment scale selected on 30 paired validation episodes. Three
1152 independently initialized graph policies and parameter-matched flat policies
1153 were then evaluated on the same 100 previously unseen paired holdout repli-
1154 cations per seed. The graph and flat models contain 524,775 and 528,404

trainable parameters, respectively, so the comparison does not give the graph policy a parameter-count advantage.

Table 11: Three-seed abrupt-regional-shift comparisons. Objective differences are in millions of modeled weighted-objective units. Each comparison contains 300 paired holdout outcomes.

Comparison	ΔJ_{tot} (10^6 units)	Relative gap	Paired 95% CI	Δ completion	Δ patients lost
Network AFR-GCN-DDPG vs. MDL-2	-25.39	-1.1337%	[-29.76, -20.95]	+0.001357	-0.70
Network AFR-GCN-DDPG vs. AFR-Flat-DDPG	-25.01	-1.1166%	[-29.35, -20.61]	+0.002836	-9.86

All three seed-level objective means favor the graph controller. Relative to MDL-2, the completion-service 95% interval is $[0.000240, 0.002458]$ and the patients-lost interval is $[-6.42, 5.07]$; aggregate patient-loss noninferiority therefore passes, but the strict per-seed clinical gate passes for two of three seeds. Relative to the matched flat policy, the completion interval is $[0.001766, 0.003898]$ and the patients-lost interval is $[-15.24, -4.31]$. This is the current statistically significant evidence for a graph-specific benefit: it appears when the action space includes geographically structured edge decisions, not when the actor modifies replenishment alone.

Table 12 explains why the total-objective percentage is smaller than the change in the component directly affected by the policy. MDL-2’s mean objective is 2.2399 billion units. Bioreactor shortage, patient loss, and expiry account for 44.10%, 43.53%, and 8.71%, respectively. The graph controller reduces the bioreactor-shortage component by 25.42 million units, or 2.57% of that component, which becomes a 1.13% improvement when divided by the complete objective. The component percentage is a mechanism measure; the complete-objective percentage remains the headline result.

This positive network checkpoint is driven primarily by advantage-filtered

Table 12: Cost decomposition for MDL-2 and Network AFR-GCN-DDPG in the abrupt-regional-shift holdout. The final column is the within-component percentage change. Components are grouped without changing the total-objective denominator used in Table 11.

Component	MDL-2 mean (10^6 units)	Share of total	AFR change (10^6 units)	Component change
Bioreactor shortage	987.891	44.10%	-25.420	-2.57%
Patient loss	975.120	43.53%	-0.352	-0.04%
Expiry	195.008	8.71%	-0.084	-0.04%
Urgency	6.118	0.27%	+0.641	+10.48%
Other operating components	75.742	3.38%	-0.178	-0.23%
Total weighted objective	2239.879	100.00%	-25.393	-1.13%

1175 teacher distillation. Earlier unconstrained offline DDPG and TD3 updates
 1176 weakened the policy or violated patient safeguards. The experiment therefore
 1177 establishes value from the graph-aware AFR controller, but it does not by
 1178 itself establish that online DDPG updates generated that value.

1179 5.6.2. Conservative online-TD3 attribution screen

1180 To test whether actual actor-critic updates add value beyond the frozen
 1181 pretrained controller, we initialized matched graph and flat residual TD3
 1182 policies from their corresponding pretrained checkpoints and ran 100 online
 1183 episodes for each of three seeds. Every run completed nonzero critic and actor
 1184 updates with finite diagnostics. Final and frozen checkpoints were evaluated
 1185 on identical 100-replication CRN holdouts.

Table 13: Final conservative TD3 outcomes under geographic regional demand drift. Values aggregate three training seeds and 100 paired holdout replications per seed.

Method	Objective (10^9 units)	Completion	Patients lost	Mfg. ineligibility
MDL-2	2.458807	0.490786	2211.86	0.073077
Final AFR-Flat-TD3	2.460346	0.490433	2213.75	0.073209
Final AFR-GCN-TD3	2.456263	0.491636	2206.70	0.072685

Table 14: Paired final-checkpoint TD3 comparisons. Objective differences are in millions of modeled weighted-objective units.

Comparison	ΔJ_{tot}	Relative gap	Paired 95% CI	Δ completion	Δ lost	Δ mfg. inelig.
Final AFR-GCN-TD3 vs. MDL-2	-2.544	-0.103459%	$[-3.872, -1.131]$	+0.000850	-5.16	-0.000392
Final AFR-Flat-TD3 vs. MDL-2	+1.539	+0.062595%	$[+0.807, +2.387]$	-0.000353	+1.89	+0.000132
Final AFR-GCN-TD3 vs. AFR-Flat-TD3	-4.083	-0.165951%	$[-5.633, -2.436]$	+0.001203	-7.05	-0.000524

The final graph policy improves on MDL-2 for each seed (-0.051936% , -0.139249% , and -0.119193%), deploys nonzero residuals in 73.50%, 47.63%, and 67.25% of evaluated decisions, and passes all pre-specified clinical non-inferiority checks. The matched flat policies are more expensive than MDL-2 and fail clinical noninferiority for all three seeds. The final graph-versus-flat comparison is therefore statistically and clinically favorable.

The online-learning attribution is more limited. The pooled final-minus-frozen graph difference is -0.350 million units, with 95% interval $[-1.032, 0.155]$ million, and the GCN-minus-flat difference-in-differences is -0.485 million units, with interval $[-1.489, 0.173]$ million. Both point estimates favor online TD3 updates, but both intervals include zero. Thus the final graph controller outperforms MDL-2 and matched flat TD3 in this regional screen, while the independent incremental benefit of online TD3 over frozen pretraining is not yet established. A five-scenario zero-shot evaluation and additional training seeds are reserved for confirmation rather than being reported as completed evidence here.

6. Discussion and Conclusions

This study develops a patient-condition and geography-aware simulation of a 20-clinic PRM network, including identity-preserving pre-manufacturing

specimen routing, and a residual GCN-DDPG controller that learns around a strong MDL-2 anchor. Advantage-filtered distillation identifies sparse state-dependent corrections, while checkpoint selection, residual-scale calibration, and fallback prevent an unsupported correction from being deployed. This structure preserves the interpretability of the anchor while allowing the graph actor to respond to patient pressure, transfer pipelines, regional disruptions, and demand-prior drift.

The numerical findings currently summarized below come from the earlier no-routing configurations and should be interpreted as controller evidence, not as evidence for or against specimen routing. The calibrated patient process places both ISO and MDL-2 inside a clinically interpretable manufacturing-period ineligibility band, and the weighted objective no longer rewards early patient loss. Under demand-prior drift, the formal five-seed result shows that AFR-GCN-DDPG improves on MDL-2 with a two-level confidence interval fully below zero while weakly improving all reported patient-facing metrics. The effect is small, and a matched flat residual policy also improves on MDL-2; consequently, this experiment supports anchored residual learning but not an independent graph advantage. The matched backbone comparison favors DDPG over TD3 for this replenishment-residual problem, which supports DDPG as the canonical proposed instantiation without implying that it is universally superior.

The graph-specific result emerges after the residual action is aligned with the network representation. Under an abrupt regional demand shift, Network AFR-GCN-DDPG significantly improves on both MDL-2 and a parameter-matched flat policy while improving completion service. The cost decompo-

sition shows that the controller primarily reduces bioreactor-shortage penalties; reporting this component change alongside the full-objective percentage explains the mechanism without changing the headline denominator. A separate conservative online-TD3 screen likewise yields a final graph policy that outperforms MDL-2 and matched flat TD3 while satisfying clinical noninferiority. However, the pooled final-versus-frozen and graph-minus-flat difference-in-differences intervals both cross zero. We therefore distinguish the supported claim that the deployed graph-aware AFR controller creates value from the unresolved claim that online actor-critic updates independently create that value.

Several limitations remain. The formal demand-prior-drift DDPG experiment has five training seeds, but the network-action DDPG and conservative TD3 studies currently have three. The abrupt-shift graph checkpoint is dominated by advantage-filtered teacher distillation, and cross-scenario zero-shot robustness is not yet established. The full disruption-rate, robust/rolling MDL-2, pure-policy, SAC, and PPO comparisons still require a common locked budget. The clinical calibration target is literature aligned but is not fitted to patient-level data from a single indication, so sensitivity bands remain necessary. The patient-loss, expiry, urgency, and shortage coefficients are modeled decision weights rather than a complete accounting valuation; cost-weight sensitivity and external economic calibration are required before interpreting objective differences as realized monetary savings. External clinical data will also be needed to validate patient-risk, demand, disruption, and transfer-time assumptions before operational use. The routing-capable environment has passed implementation-level identity, conservation, lifecycle,

1255 and CRN tests, and routing-enabled teacher and development screens have
1256 been completed. The untouched-seed routing-primary GCN/flat confirma-
1257 tion and final holdout remain pending. Accordingly, this draft treats the
1258 current routing screens as development evidence, makes no confirmatory em-
1259 pirical claim about the value of routing, and does not reinterpret any legacy
1260 no-routing checkpoint as routing-enabled evidence.

1261 **Future Work** This study motivates several extensions. Patient condi-
1262 tion can be represented with longitudinal clinical covariates rather than the
1263 current eligibility and survival-risk summary. Raw-material quality, prod-
1264 uct quality, process failures, machine breakdowns, and operator availabil-
1265 ity can be added to the simulator. The immediate empirical priority is to
1266 freeze the current routing-primary DDPG candidate, complete matched GCN
1267 and flat confirmation on five untouched seeds, and evaluate both final and
1268 frozen-pretraining controls against routing MDL-2 under the same CRN pro-
1269 tocol. Training begins with 100 episodes and is extended uniformly to 200
1270 or, only when justified by the still-sealed development learning curves, 300
1271 episodes. No-routing training is reserved for a separately justified supple-
1272 mentary ablation rather than the default experiment. If online-learning at-
1273 tribution remains inconclusive after this independent confirmation, the next
1274 candidate should change the actor-critic mechanism (for example, a matched
1275 TD3 study) rather than repeatedly tuning DDPG on the same development
1276 stream. Longer-term experiments should test unseen clinic networks and
1277 transfer disruptions and should integrate production scheduling, product
1278 preparation and distribution, and treatment operations.

1279 **Data Availability**

1280 The data used in this study are generated from a simulation model.
1281 The simulation parameters and experimental settings are described in the
1282 manuscript. Code and additional materials are available from the corre-
1283 sponding author upon reasonable request.

1284 **Declaration of Competing Interest**

1285 The author declares that there is no known competing financial interest or
1286 personal relationship that could have appeared to influence the work reported
1287 in this paper.

1288 **References**

- 1289 [1] K. Wang et al. “A simulation-based comparison of centralized and
1290 point-of-care supply chain strategies for autologous cell therapy”. In:
1291 *Cytotherapy* 25.12 (Dec. 2023). Epub 2023 Sep 22, pp. 1370–1379. DOI:
1292 10.1016/j.jcyt.2023.08.007.
- 1293 [2] Hao Zheng et al. *Opportunities of Hybrid Model-Based Reinforcement*
1294 *Learning for Cell Therapy Manufacturing Process Control*. 2022. DOI:
1295 10.48550/arXiv.2201.03116. arXiv: 2201.03116 [cs.LG]. URL:
1296 <https://arxiv.org/abs/2201.03116>.
- 1297 [3] Melanie Ayala Ceja et al. “CAR-T Cell Manufacturing: Major Process
1298 Parameters and Next-Generation Strategies”. In: *Journal of Experi-*
1299 *mental Medicine* 221.2 (2024). DOI: 10.1084/jem.20230903. URL:
1300 <https://doi.org/10.1084/jem.20230903>.

- 1301 [4] Stephen Tully et al. “Impact of Increasing Wait Times on Overall Mor-
1302 tality of Chimeric Antigen Receptor T-Cell Therapy in Large B-Cell
1303 Lymphoma: A Discrete Event Simulation Model”. In: *JCO Clinical*
1304 *Cancer Informatics* 3 (2019). DOI: 10 . 1200 / CCI . 19 . 00086. URL:
1305 <https://doi.org/10.1200/CCI.19.00086>.
- 1306 [5] Grand View Research. *CAR T-Cell Therapy Market Size, Share &*
1307 *Trends Analysis Report*. Market research report. 2024. URL: [https :](https://www.grandviewresearch.com/industry-analysis/car-t-cell-therapy-market-report)
1308 [//www.grandviewresearch.com/industry-analysis/car-t-cell-](https://www.grandviewresearch.com/industry-analysis/car-t-cell-therapy-market-report)
1309 [therapy-market-report](https://www.grandviewresearch.com/industry-analysis/car-t-cell-therapy-market-report).
- 1310 [6] Junxuan Li and Chelsea C. White. “Capacity planning in a decentral-
1311 ized autologous cell therapy manufacturing network for low-cost re-
1312 silience”. In: *Flexible Services and Manufacturing Journal* 35.2 (2023),
1313 pp. 295–319. ISSN: 1936-6590. DOI: 10 . 1007 / s10696 - 022 - 09475 - 6.
1314 URL: <https://doi.org/10.1007/s10696-022-09475-6>.
- 1315 [7] Chin-Yuan Tseng et al. “Deep reinforcement learning approach for dy-
1316 namic capacity planning in decentralized regenerative medicine supply
1317 chains”. In: *International Journal of Production Research* 63.2 (2025),
1318 pp. 555–570. DOI: 10 . 1080 / 00207543 . 2023 . 2262043. URL: [https :](https://doi.org/10.1080/00207543.2023.2262043)
1319 [//doi.org/10.1080/00207543.2023.2262043](https://doi.org/10.1080/00207543.2023.2262043).
- 1320 [8] Timothy P. Lillicrap et al. “Continuous Control with Deep Reinforce-
1321 ment Learning”. In: *International Conference on Learning Representa-*
1322 *tions (ICLR)*. 2016. URL: <https://arxiv.org/abs/1509.02971>.
- 1323 [9] Scott Fujimoto, Herke van Hoof, and David Meger. “Addressing Func-
1324 tion Approximation Error in Actor-Critic Methods”. In: *International*

- 1325 *Conference on Machine Learning (ICML)*. 2018. URL: <https://arxiv.org/abs/1802.09477>.
1326
- 1327 [10] Tuomas Haarnoja et al. “Soft Actor-Critic: Off-Policy Maximum En-
1328 tropy Deep Reinforcement Learning with a Stochastic Actor”. In: *Inter-
1329 national Conference on Machine Learning (ICML)*. 2018. URL: <https://arxiv.org/abs/1801.01290>.
1330
- 1331 [11] John Schulman et al. “Proximal Policy Optimization Algorithms”. In:
1332 *arXiv preprint arXiv:1707.06347* (2017). URL: <https://arxiv.org/abs/1707.06347>.
1333
- 1334 [12] Thomas N. Kipf and Max Welling. “Semi-Supervised Classification
1335 with Graph Convolutional Networks”. In: *International Conference on
1336 Learning Representations (ICLR)*. 2017. URL: <https://arxiv.org/abs/1609.02907>.
1337
- 1338 [13] Niki Kotecha and Antonio del Rio Chanona. “Leveraging Graph Neural
1339 Networks and Multi-Agent Reinforcement Learning for Inventory Con-
1340 trol in Supply Chains”. In: *Computers & Chemical Engineering* 199
1341 (2025), p. 109111. DOI: 10.1016/j.compchemeng.2025.109111. URL:
1342 <https://doi.org/10.1016/j.compchemeng.2025.109111>.
- 1343 [14] Serina Chang et al. “Learning Production Functions for Supply Chains
1344 with Graph Neural Networks”. In: *Proceedings of the AAAI Conference
1345 on Artificial Intelligence*. AI for Social Impact Track; arXiv:2407.18772.
1346 2025. DOI: 10.48550/arXiv.2407.18772. URL: <https://arxiv.org/abs/2407.18772>.
1347

- [15] Hyun-Il Ahn et al. *GNN-Based Probabilistic Supply and Inventory Predictions in Supply Chain Networks*. 2024. DOI: 10.48550/arXiv.2404.07523. arXiv: 2404.07523 [cs.LG]. URL: <https://arxiv.org/abs/2404.07523>.
- [16] Hyun-Il Ahn et al. *Generative Probabilistic Planning for Optimizing Supply Chain Networks*. 2024. DOI: 10.48550/arXiv.2404.07511. arXiv: 2404.07511 [cs.LG]. URL: <https://arxiv.org/abs/2404.07511>.
- [17] Ziheng Liu et al. “Graph Neural Network Meets Multi-Agent Reinforcement Learning: Fundamentals, Applications, and Future Directions”. In: *arXiv preprint arXiv:2404.04898* (2024). URL: <https://arxiv.org/abs/2404.04898>.
- [18] Mohamed Mousa et al. *An Analysis of Multi-Agent Reinforcement Learning for Decentralized Inventory Control Systems*. 2023. DOI: 10.48550/arXiv.2307.11432. arXiv: 2307.11432 [cs.LG]. URL: <https://arxiv.org/abs/2307.11432>.
- [19] Jaime Sabal Bermudez, Antonio del Rio Chanona, and Calvin Tsay. *Distributional Constrained Reinforcement Learning for Supply Chain Optimization*. 2023. DOI: 10.48550/arXiv.2302.01727. arXiv: 2302.01727 [cs.LG]. URL: <https://arxiv.org/abs/2302.01727>.
- [20] Carson Eisenach et al. *Neural Coordination and Capacity Control for Inventory Management*. 2024. DOI: 10.48550/arXiv.2410.02817. arXiv: 2410.02817 [cs.LG]. URL: <https://arxiv.org/abs/2410.02817>.

- 1372 [21] Francesco Stranieri et al. *Classical and Deep Reinforcement Learning*
1373 *Inventory Control Policies for Pharmaceutical Supply Chains with Per-*
1374 *ishability and Non-Stationarity*. 2025. DOI: 10.48550/arXiv.2501.
1375 10895. arXiv: 2501.10895 [cs.LG]. URL: [https://arxiv.org/abs/](https://arxiv.org/abs/2501.10895)
1376 2501.10895.
- 1377 [22] Peter Henderson et al. “Deep Reinforcement Learning That Matters”.
1378 In: *Proceedings of the AAAI Conference on Artificial Intelligence*. 2018.
1379 DOI: 10.48550/arXiv.1709.06560. URL: [https://arxiv.org/abs/](https://arxiv.org/abs/1709.06560)
1380 1709.06560.
- 1381 [23] Rishabh Agarwal et al. “Deep Reinforcement Learning at the Edge of
1382 the Statistical Precipice”. In: *Advances in Neural Information Process-*
1383 *ing Systems*. 2021. DOI: 10.48550/arXiv.2108.13264. URL: [https:](https://arxiv.org/abs/2108.13264)
1384 [//arxiv.org/abs/2108.13264](https://arxiv.org/abs/2108.13264).
- 1385 [24] Chin-Yuan Tseng et al. “Advancing autologous CAR T-cell therapy
1386 through real-time patient health data integration: a simulation-based
1387 approach”. In: *Cytotherapy* 26.10 (2024), pp. 1152–1162.
- 1388 [25] Ilaria Giannoccaro and Pierpaolo Pontrandolfo. “Inventory manage-
1389 ment in supply chains: a reinforcement learning approach”. In: *Interna-*
1390 *tional Journal of Production Economics* 78.2 (2002), pp. 153–161. ISSN:
1391 0925-5273. DOI: [https://doi.org/10.1016/S0925-5273\(00\)00156-](https://doi.org/10.1016/S0925-5273(00)00156-0)
1392 0. URL: [https://www.sciencedirect.com/science/article/pii/](https://www.sciencedirect.com/science/article/pii/S0925527300001560)
1393 S0925527300001560.
- 1394 [26] Ahmet Kara and Ibrahim Dogan. “Reinforcement learning approaches
1395 for specifying ordering policies of perishable inventory systems”. In:

- 1396 *Expert Systems with Applications* 91 (2018), pp. 150–158. ISSN: 0957-
1397 4174. DOI: <https://doi.org/10.1016/j.eswa.2017.08.046>.
1398 URL: [https://www.sciencedirect.com/science/article/pii/](https://www.sciencedirect.com/science/article/pii/S0957417417305900)
1399 [S0957417417305900](https://www.sciencedirect.com/science/article/pii/S0957417417305900).
- 1400 [27] Afshin Oroojlooyjadid et al. “A Deep Q-Network for the Beer Game:
1401 Deep Reinforcement Learning for Inventory Optimization”. In: *Man-*
1402 *ufacturing & Service Operations Management* 24.1 (2022). DOI: 10 .
1403 1287/msom.2020.0939.
- 1404 [28] Joren Gijsbrechts et al. “Can Deep Reinforcement Learning Improve
1405 Inventory Management? Performance on Lost Sales, Dual-Sourcing,
1406 and Multi-Echelon Problems”. In: *Manufacturing & Service Operations*
1407 *Management* 24.3 (2022), pp. 1349–1368. DOI: 10.1287/msom.2021 .
1408 1064. URL: <https://doi.org/10.1287/msom.2021.1064>.
- 1409 [29] Robert N. Boute et al. “Deep Reinforcement Learning for Inventory
1410 Control: A Roadmap”. In: *European Journal of Operational Research*
1411 298.2 (2022), pp. 401–412. DOI: 10.1016/j.ejor.2021.07.016. URL:
1412 <https://doi.org/10.1016/j.ejor.2021.07.016>.
- 1413 [30] Bram J. De Moor, Joren Gijsbrechts, and Robert N. Boute. “Reward
1414 Shaping to Improve the Performance of Deep Reinforcement Learning
1415 in Perishable Inventory Management”. In: *European Journal of Oper-*
1416 *ational Research* 301.2 (2022), pp. 535–545. DOI: 10.1016/j.ejor .
1417 2021.10.045. URL: [https://doi.org/10.1016/j.ejor.2021.10.](https://doi.org/10.1016/j.ejor.2021.10.045)
1418 [045](https://doi.org/10.1016/j.ejor.2021.10.045).

- 1419 [31] Francesco Stranieri, Edoardo Fadda, and Fabio Stella. “Combining
1420 Deep Reinforcement Learning and Multi-Stage Stochastic Program-
1421 ming to Address the Supply Chain Inventory Management Problem”.
1422 In: *International Journal of Production Economics* 268 (2024), p. 109099.
1423 DOI: 10.1016/j.ijpe.2023.109099. URL: <https://doi.org/10.1016/j.ijpe.2023.109099>.
1424
- 1425 [32] Henri Dehaybe, Daniele Catanzaro, and Philippe Chevalier. “Deep Re-
1426 inforcement Learning for Inventory Optimization with Non-Stationary
1427 Uncertain Demand”. In: *European Journal of Operational Research*
1428 314.2 (2024), pp. 433–445. DOI: 10.1016/j.ejor.2023.10.007.
1429 URL: <https://doi.org/10.1016/j.ejor.2023.10.007>.
- 1430 [33] Tobias Johannink et al. “Residual Reinforcement Learning for Robot
1431 Control”. In: *2019 International Conference on Robotics and Automa-
1432 tion (ICRA)*. 2019, pp. 6023–6029. DOI: 10.1109/ICRA.2019.8794127.
1433 URL: <https://doi.org/10.1109/ICRA.2019.8794127>.
- 1434 [34] Tom Staessens, Tom Lefebvre, and Guillaume Crevecoeur. “Adaptive
1435 Control of a Mechatronic System Using Constrained Residual Rein-
1436 forcement Learning”. In: *IEEE Transactions on Industrial Electronics*
1437 69.10 (2022), pp. 10447–10456. DOI: 10.1109/TIE.2022.3144565.
1438 URL: <https://doi.org/10.1109/TIE.2022.3144565>.
- 1439 [35] Alberto Maria Metelli et al. “Safe Policy Iteration: A Monotonically
1440 Improving Approximate Policy Iteration Approach”. In: *Journal of Ma-
1441 chine Learning Research* 22.97 (2021), pp. 1–83. URL: <https://www.jmlr.org/papers/v22/19-707.html>.
1442

- 1443 [36] Tarkan Temizöz et al. “Deep Controlled Learning for Inventory Con-
1444 trol”. In: *European Journal of Operational Research* 324.1 (2025), pp. 104–
1445 117. DOI: 10.1016/j.ejor.2025.01.026. URL: [https://doi.org/](https://doi.org/10.1016/j.ejor.2025.01.026)
1446 10.1016/j.ejor.2025.01.026.
- 1447 [37] Remi Leluc et al. *MARLIM: Multi-Agent Reinforcement Learning for*
1448 *Inventory Management*. 2023. DOI: 10.48550/arXiv.2308.01649.
1449 arXiv: 2308.01649 [cs.LG]. URL: [https://arxiv.org/abs/2308.](https://arxiv.org/abs/2308.01649)
1450 01649.
- 1451 [38] William L. Hamilton, Rex Ying, and Jure Leskovec. “Inductive Repre-
1452 sentation Learning on Large Graphs”. In: *Advances in Neural Informa-*
1453 *tion Processing Systems (NeurIPS)*. 2017. URL: [https://arxiv.org/](https://arxiv.org/abs/1706.02216)
1454 [abs/1706.02216](https://arxiv.org/abs/1706.02216).
- 1455 [39] Petar Velikovi et al. “Graph Attention Networks”. In: *International*
1456 *Conference on Learning Representations (ICLR)*. 2018. URL: [https:](https://arxiv.org/abs/1710.10903)
1457 [//arxiv.org/abs/1710.10903](https://arxiv.org/abs/1710.10903).